

Adaptive Heuristic Ecosystem for Online Multi-Robot Task Allocation under Failures, Mixed Stress, and Deadline Pressure

Abstract

We study online multi-robot task allocation under robot failures, mixed stress, and deadline pressure. AHE-MRTA uses Ecosystem-Driven Paradigm Selection (EDPS). EDPS is an event-triggered rule that maps five hormone states to five fixed allocation behaviours. Explicit overrides handle acute failure and deadline regimes. In all other regimes, a fixed dominance dynamic selects the behaviour. The F58 configuration extends the F45 core with cached geodesic estimated time of arrival (ETA) and a terminal, bounded load-repair step.

We evaluate allocation and execution in three separate settings: a 500-seed allocation-only campaign, a stochastic navigation proxy, and closed-loop Gazebo Harmonic/ROS 2 Nav2 experiments with 3, 5, and 10 TurtleBot3 robots. In the allocation-only campaign, F58 reaches the observed ceiling for completion and deadline violations in the tested cells. This setting therefore does not separate the leading policies. In the navigation proxy, AHE-MRTA and Consensus-DBTA are co-leaders; the only significant separation is a small mixed-stress edge for AHE-MRTA. A matched F45–F58 Gazebo comparison uses 10 common seeds for every scenario–scale cell. It meets the pre-specified promotion gate in seven of the nine cells. The configuration also increases decision latency and reduces distance balance under deadline pressure. The four-method Gazebo benchmark uses five seeds per cell. We therefore report it descriptively and do not use it to claim broad superiority. The current experiments also do not causally isolate EDPS from the other implementation differences in the execution stack. We state this limit, keep allocation-only and closed-loop evidence separate, and specify the matched selector ablation required for the remaining causal claim.

Index Terms

Multi-robot systems, task allocation, online algorithms, biomimetic optimization, ROS 2, Gazebo.

I. INTRODUCTION

Multi-robot task allocation (MRTA) decides which robot performs each task and in what order. The decision must respect robot capabilities, task requirements, and operational constraints. In an operating system, it also depends on execution state. A robot may be delayed, unavailable, or committed to a task whose value or feasibility has changed. Recent reviews cover centralized optimization, distributed and market-based methods, metaheuristics, and learning-based policies [1], [2]. For online MRTA, a low-cost assignment at one instant is not enough. The allocator must preserve useful decisions as tasks arrive, robots become unavailable, deadlines approach, and execution exposes new constraints.

Existing approaches make different trade-offs. Centralized formulations can jointly optimize assignment and routing when global state is available. Distributed formulations instead use local decisions and explicit communication assumptions. Bai *et al.* use group-based distributed auctions for delivery requests that appear over time and have time windows [3]. Consensus-based allocation addresses local coordination, speed, and energy use [4]. Game-theoretic clustering combines task grouping and allocation [5]. Joint allocation–path-planning formulations include timing, workload, distance, energy, and deadline objectives [6]. Other online methods adapt to changes in problem structure [7] or make decentralized energy-aware decisions [8]. Because these methods observe and change the system differently, a comparison must state the observation model, reallocation event, communication assumptions, and outcome metrics.

This letter considers online single-task-robot allocation with task arrivals, robot failures, mixed stress, and deadline pressure. The allocator acts at a task arrival, failure, deadline alarm, or replanning event. It can use only the state available at that event. Reassigning a task can reduce estimated travel cost but can also preempt execution, disrupt a feasible route, or move delay to another robot. Keeping an incumbent task can also be harmful after a failure or a sharp reduction in time slack. We therefore use completion and deadline violations as primary outcomes. We report delay, workload balance, recovery, redispaches, preemptions, and decision latency as trade-offs. A strong score in a navigation-independent allocator does not by itself demonstrate closed-loop navigation or recovery performance.

Our method, AHE-MRTA, uses Ecosystem-Driven Paradigm Selection (EDPS). EDPS is an interpretable event-triggered selector, not a new optimizer at every event. Four observable signals—task density, robot availability, deadline pressure, and recent failure rate—update five hormone states. The selected state activates one fixed behaviour: spatial greedy, priority-first, deadline-oriented temporal, commit-once stability-oriented, or orphan-first recovery allocation. Acute failure and deadline regimes have explicit overrides. A fixed dominance dynamic chooses the behaviour in all other cases. The contribution is therefore a transparent rule for switching between known policy families. It does not claim that one family is always best or that the hormone metaphor alone creates a new solver.

The paper makes four bounded contributions:

- 1) It specifies Ecosystem-Driven Paradigm Selection (EDPS), an event-triggered, interpretable policy coupling fixed context signals to fixed MRTA behaviours.
- 2) It records decision and execution quantities needed to distinguish allocation quality from navigation and recovery effects.
- 3) It evaluates the policy across multiple robot-team scales in both a navigation-independent allocation-only simulator and a ROS 2/Gazebo stack.
- 4) It reports the matched F45–F58 Gazebo comparison separately, so map-aware ETA and bounded-repair evidence is not conflated with EDPS.

The evaluation has three layers: allocation-only simulation, a stochastic navigation proxy, and closed-loop Gazebo/ROS 2 execution. The separation prevents a navigation or recovery effect from being reported as allocation evidence. The F45–F58 study tests a configuration change. It does not test the causal effect of EDPS alone in the execution stack. That claim requires a matched Gazebo selector ablation. Stating this boundary makes the comparison between the adaptive selector and fixed MRTA baselines reproducible.

II. PROBLEM FORMULATION

Event-driven state. Let $\mathcal{R} = \{r_1, \dots, r_n\}$ be the robot set. A requested task τ is described by

$$\xi_\tau = (p_\tau, t_\tau^a, d_\tau, \pi_\tau, c_\tau),$$

where $p_\tau \in \mathbb{R}^2$ is the goal position, t_τ^a is its activation time, $d_\tau \in \mathbb{R}_{\geq 0} \cup \{\infty\}$ is its deadline, $\pi_\tau \in \{1, 2, 3\}$ is its priority, and c_τ is its capability requirement. At event time t_k , $\mathcal{T}_k$ contains every activated task that is not completed or irreversibly cancelled. Robot r has position p_r^k , health state $\phi_r^k \in \{\text{alive}, \text{failed}\}$, navigation status, and an ordered queue q_r^k . The queue contains tasks assigned but not completed, including an in-flight task when one exists.

An allocation event is a task arrival, a robot failure, a deadline alarm, or the replanning timer. Let $\mathcal{R}_k^{\text{av}}$ denote robots that are alive, not navigation-stuck, and have remaining queue capacity. Let $\mathcal{I}_k \subseteq \mathcal{R} \times \mathcal{T}_k$ be the set of healthy in-flight robot–task pairs. These pairs are locked: a new allocation event may change future queue entries but does not cancel an executing Nav2 goal.

Feasible event-level decision. The binary variable $x_{r\tau}^k = 1$ means that task τ belongs to robot r 's planned queue after event t_k . The feasible set is

$$\mathcal{X}_k = \{x \in \{0, 1\}^{n \times |\mathcal{T}_k|} : \sum_{r \in \mathcal{R}} x_{r\tau}^k \leq 1, \quad \forall \tau \in \mathcal{T}_k, \quad (1)$$

$$\sum_{\tau \in \mathcal{T}_k} x_{r\tau}^k \leq Q, \quad \forall r \in \mathcal{R}, \quad (2)$$

$$x_{r\tau}^k = 1 \Rightarrow (c_\tau \subseteq \text{cap}(r)), \quad \forall r, \tau, \quad (3)$$

$$x_{r\tau}^k = x_{r\tau}^{k-1}, \quad \forall (r, \tau) \in \mathcal{I}_k. \quad (4)$$

The first constraint gives a task one owner, the second limits queue length, and the third enforces capability compatibility. The final constraint is the in-flight safety lock used in the execution stack. Failed or navigation-stuck robots cannot receive a new task; their unfinished tasks become eligible for redispach.

Arrival prediction and admissibility. For a candidate pair (r, τ) , let $\hat{a}_{r\tau}^k$ be the predicted completion-arrival time obtained by traversing q_r^k and then travelling to p_τ . The predictor uses the same route metric as the active configuration: F45 uses its core distance estimate, while F58 uses the cached geodesic ETA in Eq. (19). A pair with no route is infeasible. For a finite deadline, the admission rule is

$$\hat{a}_{r\tau}^k > d_\tau + s_{r\tau}^k \implies (r, \tau) \notin \mathcal{F}_k, \quad (5)$$

where $\mathcal{F}_k$ is the eligible-pair set. The pair-specific allowance $s_{r\tau}^k$ is zero for new tasks in the nominal configuration. A bounded positive allowance is available only to old rescue tasks and can be widened in failure mode. Thus, the allocator rejects a pair that is incompatible, unavailable, unreachable, or later than the allowance for that task state.

Event-level surrogate. Minimizing non-negative cost over $\mathcal{X}_k$ alone would admit the trivial all-zero assignment because the ownership constraints are upper bounds. The implementation therefore uses a cardinality-first rule. After masking infeasible pairs and preserving locked incumbents, define

$$\mathcal{X}_k^* = \arg \max_{x \in \mathcal{X}_k} \sum_{r \in \mathcal{R}} \sum_{\tau \in \mathcal{T}_k} x_{r\tau}^k. \quad (6)$$

Among these maximum-cardinality event plans, the selected paradigm minimizes its shaped cost:

$$x^k \in \arg \min_{x \in \mathcal{X}_k^*} \sum_{r \in \mathcal{R}} \sum_{\tau \in \mathcal{T}_k} C_{r\tau}^k x_{r\tau}^k, \quad C_{r\tau}^k = \infty \text{ if } (r, \tau) \notin \mathcal{F}_k. \quad (7)$$

Repeated LSA passes implement this lexicographic rule; tasks for which no feasible capacity remains stay open for a later event. The active paradigm then sets queue order (Section III-E). This is an online event-level surrogate, not an offline optimum for the complete mission.

Cost and assignment stability. The implementation evaluates seven normalized features: arrival time $A_{r\tau}^k$, priority penalty $P_\tau = (4 - \pi_\tau)/3$, battery risk B_r^k , relative load L_r^k , failure risk F_r^k , deadline urgency U_τ^k , and navigation/recovery risk R_r^k . For a feasible pair, the common cost is

$$C_{r\tau}^k = w_d A_{r\tau}^k + w_p P_\tau + w_b B_r^k + w_l L_r^k + w_f F_r^k + w_t U_\tau^k + w_r R_r^k + \Psi_{r\tau}^k. \quad (8)$$

Here $A_{r\tau}^k = \min(1, \hat{a}_{r\tau}^k/A_{\max})$ and the other base features are dimensionless. The correction $\Psi_{r\tau}^k$ collects the deadline and critical-battery penalties, deadline-capability and bid bonuses, incumbent sticky bonus, reassignment penalty, and capacity penalty. This form avoids adding seconds directly to normalized features. EDPS supplies w ; the active paradigm can also shape task order and eligibility. A safe incumbent retains base urgency, whereas an unassigned or at-risk task receives nonlinear urgency growth.

Run-level outcomes and timing contract. The run-level objective is to maximize completion and minimize deadline violations without hiding unfinished tasks. Let $\mathcal{T}^{\text{done}} \subseteq \mathcal{T}^{\text{req}}$ be the tasks completed by $T_{\max}$. Formally,

$$\max \text{CR} = \frac{|\mathcal{T}^{\text{done}}|}{|\mathcal{T}^{\text{req}}|}, \quad \min \text{DVR} = \frac{|\{\tau : d_\tau < \infty \wedge (t_\tau^c > d_\tau \vee \tau \notin \mathcal{T}^{\text{done}})\}|}{\max(1, |\{\tau : d_\tau < \infty\}|)}. \quad (9)$$

The remaining reported quantities—delay, makespan, recovery time, workload balance, redispaches, preemptions, decision latency, messages, and distance—describe the cost of these outcomes. Delay and DVR are evaluated on all requested tasks; an unfinished task is right-censored at $T_{\max}$ for delay and is a deadline miss for DVR (Section IV-G).

For the deployed AHE cycle, the decision-time contract is $\ell_k = t_k^{\text{publish}} - t_k \leq 50$ ms. Baseline decision latencies are reported as measured comparison outcomes rather than being silently clipped to this AHE service target. This distinction preserves the real-time requirement while making latency trade-offs visible.

III. AHE-MRTA: ECOSYSTEM-DRIVEN PARADIGM SELECTION

Design principle and event-triggered architecture.

AHE-MRTA is not a new global MRTA solver; it is an event-triggered *selection hyper-heuristic* operating over a fixed library of low-level allocation behaviours [13]. At an allocation event, the fleet state, open task pool, and previous queues are observed; the selector first updates its context and internal state, then activates one of five paradigms, and finally publishes robot-specific queues. Adaptation thus occurs in two distinct decision layers: (i) the allocation behaviour to run and (ii) the weights assigned to the channels of the common cost function. The selector does not run a new inter-robot negotiation protocol; the central manager publishes only each robot's own queue.

The persistent selector state comprises a five-dimensional dominance vector $D_k \in \mathbb{R}_{\geq 0}^5$, the last accepted paradigm, and the remaining dwell counter for that paradigm. The allocator additionally retains in-flight goals, previous task–robot ownership, and an orphan-task pool. This second group is not selector memory; it is needed to enforce the in-flight lock and reassignment safety rules of Section II. Figure 1 summarizes the information flow from an event to queue publication.

A. Context representation

At each allocation event t_k , the observable system state is compressed into the four-dimensional context vector

$$c_k = [c_{1,k}, c_{2,k}, c_{3,k}, c_{4,k}]^\top \in [0, 1]^4. \quad (10)$$

Let $n = |\mathcal{R}|$ be the fleet size and $m_k = |\mathcal{T}(t_k)|$ the number of open tasks. Let $\mathcal{R}_k^{\text{av}}$ denote alive, non-stuck robots with queue capacity, and $\mathcal{R}_k^{\text{f}}$ failed or stuck robots. The components are

$$c_{1,k} = \min\left(1, \frac{m_k}{\max(1, n)}\right), \quad \text{task density}, \quad (11)$$

$$c_{2,k} = \frac{|\mathcal{R}_k^{\text{av}}|}{\max(1, n)}, \quad \text{robot availability}, \quad (12)$$

$$c_{3,k} = \frac{|\{\tau \in \mathcal{T}(t_k) : d_\tau - t_k \leq \Delta_d\}|}{\max(1, m_k)}, \quad \text{deadline pressure}, \quad (13)$$

$$c_{4,k} = \frac{|\mathcal{R}_k^{\text{f}}|}{\max(1, n)}, \quad \text{failure rate} \quad (14)$$

Here, $\Delta_d = 60$ s. Tasks with an infinite deadline do not enter the $c_{3,k}$ count. The first two signals describe demand–supply balance, the third the fraction of tasks with shrinking time slack, and the last the capacity loss requiring recovery. They are

TABLE I

CONTEXT PROTOTYPE VECTORS $V_i \in [0, 1]^4$ FOR EACH PARADIGM. COLUMN HEADERS: c_1 TASK-DENSITY (TD), c_2 ROBOT-AVAILABILITY (RA), c_3 DEADLINE-PRESSURE (DP), c_4 FAILURE-RATE (FR). HIGH VALUE = PARADIGM EFFECTIVE WHEN THIS CONTEXT DIMENSION IS HIGH.

Hormone	td	ra	dp	fr
H_SPATIAL	0.7	0.7	0.1	0.1
H_CRIT	0.3	0.5	0.8	0.2
H_TEMP	0.5	0.5	0.9	0.1
H_STAB	0.3	0.3	0.3	0.8
H_RECOV	0.3	0.2	0.2	0.9

TABLE II

NONZERO ENTRIES OF COOPERATION MATRIX A AND SUPPRESSION MATRIX S . ALL OTHER ENTRIES ARE ZERO.

Matrix	From	To	Value
A (cooperation)	H_CRIT	H_TEMP	0.20
	H_STAB	H_RECOV	0.20
S (suppression)	H_TEMP	H_SPATIAL	0.30

computed from the task pool and already published robot-status summaries; EDPS therefore adds no inter-robot communication burden.

Each paradigm $i \in \{1, \dots, 5\}$ has a designer-defined context prototype $V_i \in [0, 1]^4$. The prototypes are not learned class centres; they are fixed, interpretable priors on which signal combination supports the associated behaviour. Compatibility with the current context is

$$v_{i,k} = \frac{V_i^\top c_k}{\|V_i\|_2 \|c_k\|_2} \quad \text{and} \quad \mathbf{v}_k = [v_{1,k}, \dots, v_{5,k}]^\top, \quad (15)$$

with zero compatibility if either vector has zero norm. Table I gives the prototypes.

B. Dominance update and cost weights

The dominance vector is initialized uniformly and kept on the probability simplex at every event: $D_0 = \frac{1}{5}\mathbf{1}$, $D_k \succeq 0$, and $\|D_k\|_1 = 1$. Let N_k^c and N_k^f be, respectively, the numbers of completed and failed tasks between events k and $k+1$. The compatibility-scaled performance feedback and failure support are

$$\mathbf{p}_k = g_k \mathbf{v}_k, \quad g_k = \frac{N_k^c - N_k^f}{\max(1, N_k^c + N_k^f)}, \quad \mathbf{b}_k = c_{4,k} [-0.3, 0, 0, 0.4, 0.6]^\top. \quad (16)$$

Thus, a failure supports H_RECOV and H_STAB while reducing the dominance of the purely spatial behaviour. The cooperation matrix A and suppression matrix S are sparse. $A_{ij} > 0$ means that paradigm j supports paradigm i ; $S_{ij} > 0$ means that j suppresses i . Their nonzero entries are reported in Table II.

The update is

$$D_{k+1} = \text{Proj}_{\Delta^4} \left(\text{clip}_{[0,1]} [\alpha D_k + \eta A D_k - \lambda S D_k + \beta \mathbf{p}_k + \gamma \mathbf{v}_k + \delta \mathbf{b}_k] \right), \quad (17)$$

where Proj_{Δ^4} normalizes the vector onto the five-element probability simplex. The rule combines interaction-inspired winner-reinforcement and competitor-suppression terms with short-horizon performance and context compatibility [12]. It is not an online learned model and does not provide an optimality guarantee; all matrices and coefficients are fixed.

a) *Weight generation.*: EDPS also produces the seven weights of the common cost function. The channels correspond, in order, to travel time, priority, battery risk, queue load, failure risk, deadline urgency, and reassignment/recovery cost. The fixed paradigm–weight map is

$$M = \begin{bmatrix} 0.9 & 0.1 & 0.1 & 0.3 & 0.3 \\ 0.1 & 0.9 & 0.5 & 0.5 & 0.1 \\ 0.1 & 0.1 & 0.1 & 0.3 & 0.3 \\ 0.1 & 0.1 & 0.1 & 0.1 & 0.3 \\ 0.1 & 0.1 & 0.1 & 0.9 & 0.9 \\ 0.1 & 0.5 & 0.9 & 0.1 & 0.1 \\ 0.1 & 0.1 & 0.1 & 0.3 & 0.9 \end{bmatrix}. \quad (18)$$

Accordingly,

$$w_k^{\text{eco}} = \text{softmax}\left(\frac{MD_k}{T_w}\right), \quad w_k = 0.3w_0 + 0.7w_k^{\text{eco}}, \quad (19)$$

where $T_w = 0.3$. When recovery mode is active, w_k is additionally blended toward w_{rec} with $\theta_k = \min(0.80, 0.50 + 0.60c_{4,k})$. If deadline pressure exceeds 0.5 and recovery mode is inactive, the urgency channel is multiplied by 1.5. These weights are the context-dependent part of the common cost in Section II. The Gazebo execution stack has no battery model, so the battery cost component is zero; its channel nevertheless remains in the softmax vector.

b) *Feasibility and stability.*: Every paradigm applies the hard admission gate of Eq. (5) and the single-owner and in-flight constraints. For unassigned or at-risk tasks that have consumed more than 0.7 of their deadline budget, the urgency penalty is increased nonlinearly. A task expected to finish on time with its incumbent owner is exempt. Changing a previous owner requires at least a 10% improvement in estimated arrival; orphan and rescue tasks are exempt from this margin rule. These safeguards prevent event-triggered replanning from becoming gratuitous task migration.

C. F58 configuration: geodesic ETA and bounded load repair

F58 is an optional cost-oracle and post-processing configuration on top of the F45 core; it changes neither the EDPS selector nor the paradigm library. F45 uses Euclidean travel distance. F58-P0 uses cached geodesic distance d_G on the 0.55 m-inflated occupancy mask of the `obstacle_map.pgm` shared with Gazebo, using an eight-connected free-cell graph. A pair is infeasible if no route exists. For robot r with ordered queue $Q_r = (q_1, \dots, q_j)$, the route-profile ETA for candidate τ is

$$\hat{a}_{r\tau,k} = t_k + \frac{d_G(p_r, p_{q_1}) + \sum_{\ell=1}^{j-1} d_G(p_{q_\ell}, p_{q_{\ell+1}}) + d_G(p_{q_j}, p_\tau)}{v_r} + \sum_{\ell=1}^j (s_{q_\ell} + h), \quad (20)$$

which reduces to $d_G(p_r, p_\tau)$ for an empty queue. Here, s_{q_ℓ} is service time and h is the fixed allowance for queued navigation. The same route metric is used by the admission gate, paradigm costs, and repair.

After the primary plan x^0 is produced, F58-P1 considers local candidates that change the robot only for newly assigned tasks. It therefore never cancels an executing Nav2 goal. Let $M(x)$ be the predicted number of deadline misses, $D_G(x)$ total geodesic distance, $J(x)$ the projected active-robot Jain index, $B_r(x)$ accumulated and committed productive load, and $F(x)$ the maximum predicted remaining completion time. The admissible candidate set is

$$\begin{aligned} \mathcal{N}(x^0) = \{x : M(x) \leq M(x^0), \quad D_G(x) \leq (1 + \epsilon)D_G(x^0), \\ J(x) \geq J(x^0), \quad \max_r B_r(x) \leq \max_r B_r(x^0), \\ F(x) \leq F(x^0)\}, \quad \epsilon = 0.10. \end{aligned} \quad (21)$$

Within this set, candidates are ranked by first increasing $J(x)$ and then, in order, decreasing maximum load, load range, and total distance. Repair is enabled only in the terminal phase, when the remaining active-task count is at most three times the healthy-robot count. For an executing first task, $F(x)$ can use Nav2's published remaining-path length instead of a static grid query. These guards prevent F58 from accepting a visually more balanced reassignment that increases deadline misses, route cost, or remaining makespan.

The F45–F58 comparison tests the effect of map-aware ETA and bounded repair; it does not independently test the causal effect of the EDPS selector. This distinction is maintained in the interpretation of the results.

D. Paradigm dispatch and dwell hysteresis

Acute conditions do not wait for the dominance update to respond. The raw choice $\tilde{p}_k$ follows the priority order

$$\tilde{p}_k = \begin{cases} \text{H_RECOV}, & c_{4,k} > 0.05, \\ \text{H_TEMP}, & c_{4,k} \leq 0.05 \wedge c_{3,k} > 0.50, \\ \arg \max_i D_{i,k}, & \text{otherwise.} \end{cases} \quad (22)$$

Thus, recovery has priority when an event contains both failure and deadline pressure. H_TEMP is the fallback if D_k is absent or its component range is below 10^{-4} , i.e., if the state is not discriminative.

Because the raw choice can switch rapidly, a change other than H_RECOV is held for four allocation events. More specifically, if $\tilde{p}_k \neq p_{k-1}$ while the dwell counter is positive, the dispatcher returns $p_k = p_{k-1}$ and decrements the counter. Once a change is accepted, the counter is reset to $\rho = 4$. H_RECOV bypasses this hold, so a detected failure does not defer the recovery pass to the next event. This mechanism retains override reactivity while limiting ping-pong switching in normal regimes.

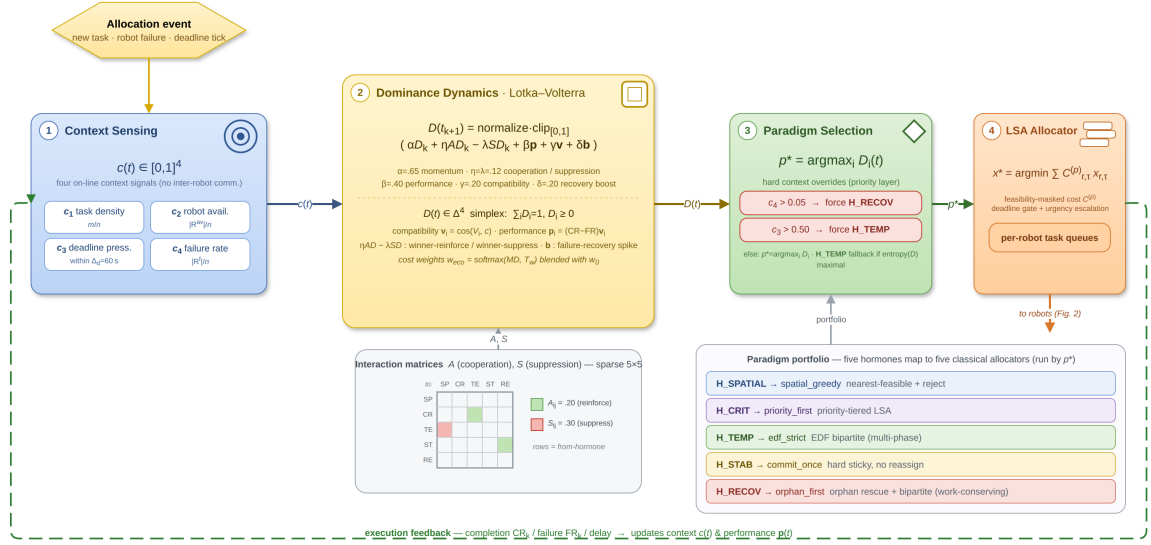


Fig. 1. Ecosystem-Driven Paradigm Selection. Context c_k updates the dominance state; failure/deadline overrides and dwell hysteresis select the active paradigm. The selected paradigm and cost weights produce robot-specific queues through feasibility-masked assignment.

TABLE III
FIVE HORMONES, FIVE ALLOCATION PARADIGMS.

Hormone	Paradigm	Inner mechanism
H_SPATIAL	spatial_greedy	nearest-feasible + reject
H_CRIT	priority_first	priority-tiered LSA
H_TEMP	edf_strict	EDF bipartite (multi-phase)
H_STAB	commit_once	hard sticky, no reassign
H_RECOV	orphan_first	orphan rescue + bipartite

E. Paradigm library

The five behaviours share the same feasible set and cost infrastructure; they differ in task-processing order, cost shaping, and treatment of incumbent ownership. Given a feasibility-masked cost matrix $C_k^{(p)}$, the common assignment kernel is

$$x_k^{(p)} \in \arg \min_{x \in \mathcal{X}_k} \sum_{r \in \mathcal{R}} \sum_{\tau \in \mathcal{T}(t_k)} C_{r\tau,k}^{(p)} x_{r\tau}. \quad (23)$$

For an infeasible robot-task pair, $C_{r\tau,k}^{(p)} = \infty$. In the implementation, linear sum assignment (LSA) is repeated across task tiers or processing phases. Equation (22) is therefore the surrogate for one assignment pass, not a claim of global optimality for the entire run.

- **H_SPATIAL / spatial_greedy:** Tasks are processed in deadline order and assigned to the eligible robot with the smallest expected arrival time. This follows the nearest-feasible rule used in metric MRTA [9], [14].
- **H_CRIT / priority_first:** Tasks are partitioned into priority tiers $\pi_\tau = 3, 2, 1$. LSA is applied first to the most critical tier and then to lower tiers, so critical tasks claim capacity first. The design follows the tiered allocation idea in priority-aware auction and consensus strategies [15], [16].
- **H_TEMP / edf_strict:** The default temporal behaviour processes tasks in increasing d_τ order and applies the hard deadline gate. The implementation uses a fast pass for near-deadline tasks, a recovery pass for any current orphans, and an LSA pass for the remaining tasks. Queue order may be improved by cheapest-insertion candidates that preserve deadline feasibility. The behaviour applies EDF to deadline-constrained MRTA [10], [17].
- **H_STAB / commit_once:** Tasks with a previous owner that remains eligible are kept with that owner. Only new tasks receive a single-pass allocation. This lowers reassignment churn but limits flexibility after a failure, making the stability trade-off explicit [7].
- **H_RECOV / orphan_first:** Tasks left by failed, stuck, or navigation-aborted robots are first redistributed among healthy robots through LSA; remaining tasks then receive temporal allocation. Recovery therefore obtains a dedicated decision pass before new work [11], [14].

Algorithm 1 AHE-MRTA Event Cycle

Require: $\mathcal{R}$, $\mathcal{T}(t_k)$, previous queues, D_k , p_{k-1} , dwell counter h_k ; fixed A, S, V, M

- 1: $c_k \leftarrow \text{CONTEXTVECTOR}(\mathcal{R}, \mathcal{T}(t_k))$
 - 2: Update the feasibility mask, orphan pool, and in-flight locks
 - 3: $\mathbf{v}_k \leftarrow (\cos(V_i, c_k))_{i=1}^5$; $\mathbf{p}_k, \mathbf{b}_k \leftarrow \text{PERFORMANCEANDFAILURESSUPPORT}()$
 - 4: $D_{k+1} \leftarrow \text{UPDATEDOMINANCE}(D_k, \mathbf{v}_k, \mathbf{p}_k, \mathbf{b}_k, A, S)$ {Eq. (16)}
 - 5: $w_k \leftarrow \text{GENERATEWEIGHTS}(D_{k+1}, c_k, M)$ {Eq. (18)}
 - 6: $\tilde{p}_k \leftarrow \text{RAWCHOICE}(c_k, D_{k+1})$ {Eq. (21)}
 - 7: $(p_k, h_{k+1}) \leftarrow \text{APPLYDWELL}(\tilde{p}_k, p_{k-1}, h_k)$
 - 8: $x_k \leftarrow \text{PARADIGM}_{p_k}(\mathcal{R}, \mathcal{T}(t_k), w_k)$ {Table III}
 - 9: **if** F58 repair is enabled **then**
 - 10: $x_k \leftarrow \text{EPSILONLOADREPAIR}(x_k, d_G, \epsilon)$
 - 11: **end if**
 - 12: $x_k \leftarrow \text{SINGLEOWNERANDINFLIGHTLOCK}(x_k)$
 - 13: Publish robot-specific queues and store (D_{k+1}, p_k, h_{k+1})
-

F. Algorithm

Algorithm 1 gives the execution order at one EDPS event. A task arrival, robot failure, deadline alarm, or replanning timer triggers the cycle. Only queues are published at the end: D_k , A , S , V , and the full cost matrix remain at the manager node.

G. Configuration

The same selector settings are used at all scenarios and fleet scales: $\alpha = 0.65$, $\beta = 0.40$, $\gamma = 0.20$, $\eta = \lambda = 0.12$, $\delta = 0.20$, $T_w = 0.3$, and $\rho = 4$. The base cost profile is $w_0 = (0.34, 0.10, 0.04, 0.16, 0.10, 0.22, 0.04)$ and the recovery profile is $w_{\text{rec}} = (0.55, 0.04, 0.03, 0.22, 0.09, 0.05, 0.02)$, in the channel order given under *Weight generation*. These quantities are neither relearned nor retuned by event or scenario.

Context, dominance, override, and dwell updates have fixed dimension and are therefore $O(1)$. The dominant cost is the LSA call within a paradigm, whose worst-case complexity for one square assignment pass is $O(\max(n, m_k)^3)$. When F58 is enabled, the geodesic cache reduces repeated goal queries and load repair is restricted to a fixed number of local move passes. Computational load thus stems from the selected paradigm and optional route/repair layers rather than from the selector itself.

IV. EXPERIMENTAL DESIGN

A. Hardware Platform

All experiments run on an Intel Core i7-11800H workstation with eight physical cores, 16 threads, a 2.30 GHz base clock, 16 GiB DDR4 RAM, Ubuntu, 24.04 LTS, and Linux, 6.17. The machine has an NVIDIA RTX, 3050, Mobile GPU, but no proprietary driver is installed. Gazebo and RViz therefore use Mesa llvmpipe software rasterization (`LIBGL_ALWAYS_SOFTWARE=1`, `GALLIUM_DRIVER=llvmpipe`). No experiment uses GPU acceleration. The reported comparison is CPU-bound.

B. Software Stack

We use **Gazebo Harmonic** (gz-sim 8.x) [18], **ROS 2 Jazzy** [19], and **Nav2** [20]. Each robot is a **TurtleBot3 Waffle Pi** [21] with a 360-ray two-dimensional lidar (5 Hz, 8 m range), a 200 Hz IMU, and a differential-drive controller ($v_{\text{max}}=0.26$ m/s, $\omega_{\text{max}}=1.82$ rad/s).

Each robot has an independent Nav2 lifecycle in its own ROS 2 namespace. The stack uses global and local costmaps with a 0.40 m inflation radius, a Regulated Pure-Pursuit controller, and a behavior-tree navigator. We provide ground-truth localization through a static `map`→`odom` transform anchored at each robot's spawn pose. This is an experimental control: it isolates allocation effects from localization error (Section VII-F). The `ros_gz_bridge` relays `/scan`, `/odom`, `/cmd_vel`, `/joint_states`, and TF for each robot. A TF relay aggregates the namespaced trees for visualization. All methods use identical Nav2 parameters.

C. Inspection Arena

The arena is a 20×20 m walled indoor mock-up. Two horizontal walls with a 1.5 m central passage create upper and lower lanes. Two rows of square pillars (0.5×0.5 m) line the side walls, and four cylindrical obstacles ($r=0.35$ m) lie in the lanes. We define 42 fixed inspection waypoints. The waypoints are obstacle-aware and at least 1 m apart. Each run samples its task set deterministically from this grid using its seed. Figure IV-C shows the arena, spawn positions, and sampled task sets.

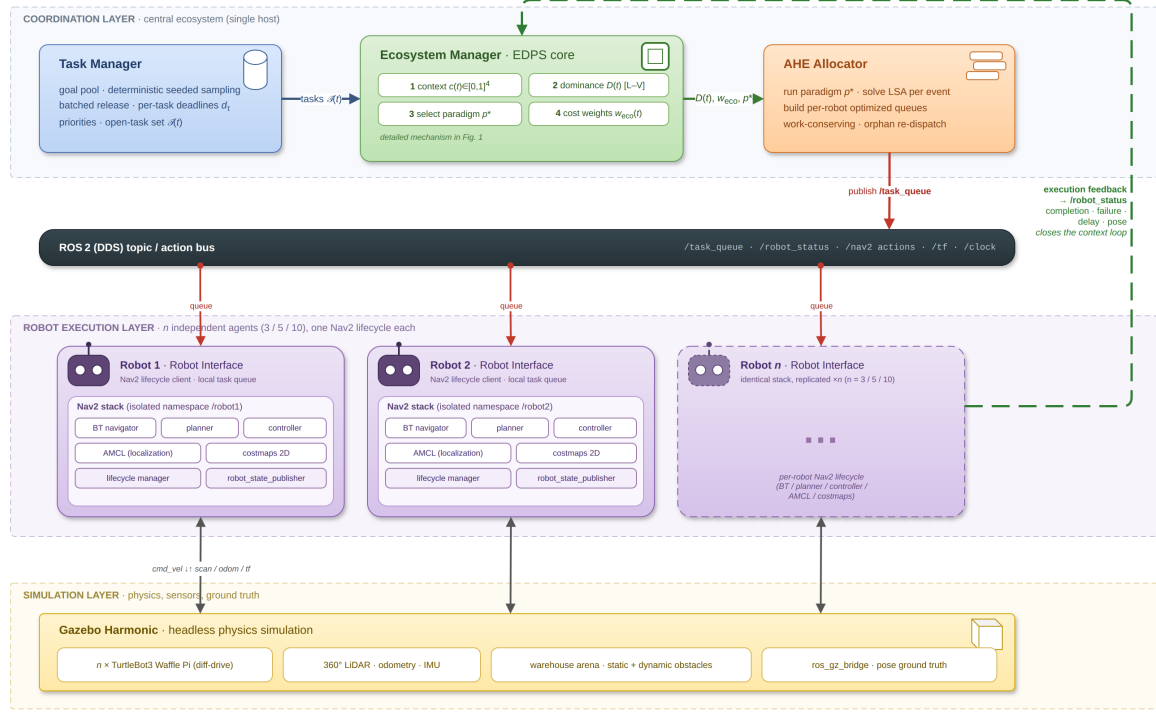


Fig. 2. System architecture. The ecosystem manager updates the hormone state, selects the active paradigm, and publishes one task queue per robot through ROS 2 topics. Each robot runs its own Nav2 lifecycle and returns completion, failure, and delay feedback to the manager.

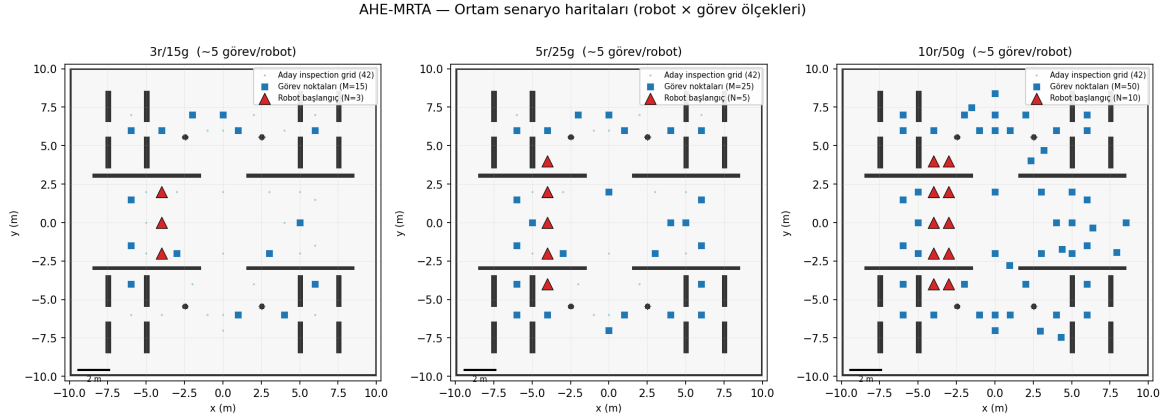


Fig. 3. Arena configurations by scale. **Left:** 3r/15t, with robots in one central column. **Center:** 5r/25t, with two spawn columns. **Right:** 10r/50t, with all five columns active. Blue stars are task goals for seed 1; colored triangles are robot spawns. All waypoints are obstacle-free.

D. Evaluation Scales

We evaluate three Gazebo scales (Table IV). Each scale contains 60 runs: 4 methods $\times$ 3 scenarios $\times$ 5 seeds.

At $N > 3$, some per-robot `odom` $\rightarrow$ `base_link` transforms arrived after costmap activation. We resolve this lifecycle race by delaying activation for robot i by $6i$ s at 3r and 5r and by $20i$ s at 10r. We also broadcast an identity transform at node startup. All scales use the same arena, obstacle map, and Nav2 parameters.

E. Stress Scenarios

The three scenarios stress different allocation requirements:

- robot_failure**: one randomly chosen robot fails at $t = 0.3 \cdot T_{\text{max}}$ (approximately 165 s at 3r). Its assigned tasks become orphaned and must be redistributed. This scenario tests reactive recovery and the orphan-rescue behaviour.
- mixed_stress**: tasks have low, medium, and high priorities in a 1:2:2 ratio. One robot fails at $t = 0.2 \cdot T_{\text{max}}$. This scenario tests adaptation to priority and failure at the same time.

TABLE IV
GAZEBO EVALUATION SCALES

Scale	Robots	Tasks	Density	Total exps
3r-low	3	9	3.0 t/r	60
3r-mid	3	15	5.0 t/r	60
3r-high	3	24	8.0 t/r	60
5r-mid	5	25	5.0 t/r	60
10r-mid	10	50	5.0 t/r	60
Total Gazebo experiments				300

- **deadline_pressure:** every task has deadline $d_\tau = t_{\text{spawn}} + 90$ s. This is about 1.3 times the Nav2 travel time to the farthest waypoint at v_{max} . The scenario tests urgency-driven dispatch and EDF ordering.

F. Baseline Methods

We compare one representative from each online-MRTA solver family represented in the AHE-MRTA library. We re-implement all baselines in the same framework. They share the event loop, ROS2 topic interface, cost inputs (arrival time, urgency, priority, and load), per-robot queue cap Q , and Nav2 client. Only the allocation policy differs. We use source-paper parameters where they are specified and keep them fixed across scenarios and scales.

a) BiG-MRTA [22] (bipartite matching): BiG-MRTA builds a bipartite robot–task graph with fixed multi-criteria edge cost and solves a maximum-weight matching at each event. It commits an assignment once and never revisits it. It is the deterministic, lowest-latency, single-paradigm comparison. Its policy avoids replanning but does not recover a task when its assigned robot fails or time slack collapses.

b) RoSTAM-EA [7] (evolutionary): RoSTAM-EA evolves candidate assignment vectors with tournament selection, crossover, and mutation at each allocation event. It dispatches the elite individual. It represents self-adaptive stochastic search. It searches a richer assignment space than one-shot matching, but its decision latency is two to three orders of magnitude higher. Its reoptimization does not encode the cause of a context change.

c) Consensus-DBTA [4] (consensus bundle): Consensus-DBTA has robots iteratively bid on task bundles and resolve conflicts in a consensus phase. It produces highly stable assignments (allocation instability ≈ 0). It represents distributed auction/consensus and is the strongest completion-rate baseline in this study. Re-bidding rounds can delay response to a failure or rapid deadline escalation.

We do not include a single-paradigm AHE ablation. Each paradigm is a classical primitive, and the corresponding baseline family approximates its single-paradigm behavior (Table III).

d) Comparison scope: The baseline implementations share the event loop, interfaces, and cost inputs, but they are operational re-implementations rather than independent replications of every source system. The four-method benchmark therefore compares these implementations in one common stack. It is not a universal ranking of the cited algorithms. A controlled Gazebo EDPS ablation against fixed paradigms is required before any closed-loop difference can be attributed to the selector itself (Section VII-F).

G. Metrics

We report completion rate (CR), deadline violation rate (DVR), makespan, all-task average delay, failure-recovery time, workload balance (Jain), task redispach rate, execution preemptions, decision latency, and message count. We also report total travel distance where applicable. CR and DVR are the primary outcomes. The other metrics describe efficiency and trade-offs.

a) All-task delay and DVR: A completed-only average rewards a policy that abandons difficult work. Distant, late, or contended tasks then disappear from the denominator. We avoid this survivorship bias by scoring delay and DVR over *all* requested tasks. Let $\mathcal{T}$ be the requested set, t_τ^a the activation time, t_τ^c the completion time, and T_{max} the run horizon. Delay for an unfinished task is right-censored at the horizon:

$$\delta_\tau = \begin{cases} t_\tau^c - t_\tau^a, & \tau \text{ completed,} \\ T_{\text{max}} - t_\tau^a, & \tau \text{ unfinished,} \end{cases} \quad \bar{\delta} = \frac{1}{|\mathcal{T}|} \sum_{\tau \in \mathcal{T}} \delta_\tau, \quad (24)$$

Any task with a deadline is a violation if it completes late *or* never completes:

$$\text{DVR} = \frac{|\{\tau : d_\tau < \infty \wedge (t_\tau^c > d_\tau \vee \tau \text{ unfinished})\}|}{|\{\tau : d_\tau < \infty\}|}. \quad (25)$$

We apply this convention to every method and scale. We also record completed-only delay and DVR. The two definitions coincide when CR=1 and differ only when a method leaves tasks unserved.

TABLE V
F58 (P0 GEODESIC ORACLE + P1R TERMINAL LOAD REPAIR) NAV2-INDEPENDENT ALLOCATION-ONLY RESULTS (100 SEEDS PER CELL). NAVIGATION SUCCEEDS DETERMINISTICALLY; DISTANCE IS THE GEODESIC PATH LENGTH ON THE SHARED INFLATED OCCUPANCY MAP.

Scale	Scenario	Fit.	CR	DVR	Jain _{active}	Jain _{dist}	Delay (s)	Distance	Decision (ms)
3r/15t	Robot failure	1.000	1.000	0.000	0.991	0.833	40.2	154.1	0.063
3r/15t	Mixed stress	1.000	1.000	0.000	0.992	0.835	40.0	153.6	0.063
3r/15t	Deadline pressure	1.000	1.000	0.000	0.988	0.965	65.8	155.5	0.061
5r/25t	Robot failure	1.000	1.000	0.000	0.986	0.841	30.8	186.5	0.086
5r/25t	Mixed stress	1.000	1.000	0.000	0.985	0.843	30.5	184.6	0.086
5r/25t	Deadline pressure	1.000	1.000	0.000	0.984	0.954	60.5	233.5	0.078
10r/50t	Robot failure	1.000	1.000	0.000	0.982	0.862	31.6	284.1	0.200
10r/50t	Mixed stress	1.000	1.000	0.000	0.985	0.861	31.4	284.1	0.201
10r/50t	Deadline pressure	1.000	1.000	0.000	0.990	0.944	57.1	432.0	0.132

H. Statistical Analysis

We use pairwise, two-sided Mann–Whitney U tests with Bonferroni correction within each scenario family (three baselines $\times$ seven metrics, or 21 tests per scenario). At the primary 5-robot scale, each cell has $n=5$ runs (one density and 25 tasks). The 3-robot scale pools three densities ($n=15$), and the 10-robot scale has $n=5$. We also report Cliff’s δ , an effect-size estimate suited to small cells. The four-method Gazebo cells with $n=5$ are descriptive. We make no cross-method superiority claim from an uncorrected or non-significant comparison. The 500-seed allocation-only campaign and stochastic navigation proxy are separate controlled tests. The matched F45–F58 Gazebo study uses paired tests with 10 common seeds per scenario and scale. Gazebo results describe behavior in the evaluated execution stack, not general deployment performance.

V. ALLOCATION-FITNESS RESULTS (NAV2-INDEPENDENT)

In the Nav2-independent setting, every navigation attempt succeeds deterministically. Injected robot failures remain part of the scenario. After each completed task, the robot pose advances to the goal. The allocator and execution oracle both use geodesic distance on the same inflated occupancy map. The metrics therefore measure allocation, queue order, and recovery after failure, rather than local-planner noise.

Table V shows that F58 has fitness=CR=1 and DVR=0 in all nine scale–scenario cells. The active-robot Jain index is 0.988–0.992 at 3r, 0.984–0.986 at 5r, and 0.982–0.990 at 10r. Decision latency is at most 0.201 ms, including the largest 10r/50t cell. During development, six 10r seeds allowed one task to execute on two robots, which inflated CR to 1.02. After adding the in-flight owner lock, we reran those seeds and the complete 10r campaign. CR is then exactly 1.000 in every F58 cell.

To separate F58 from F45, we ran a seed-matched campaign in four cells: 3r/15t, 5r/25t, and 10r/50t with seeds 1–100, plus a 5r/25t holdout with seeds 501–600. The holdout was not used to choose the repair threshold. Both arms use the same geodesic execution oracle. We use paired Wilcoxon tests with Holm correction over 10 metric families per cell–scenario. Of 120 tests, 51 are significant after Holm correction; 39 favor F58. Mean delay decreases by 0.97–2.25 s, total geodesic distance by 10–28 m, and churn decreases. The active-robot Jain index increases by +0.005–0.029 and the all-robot index by +0.006–0.027. The largest effect is at 10r/50t, where Cliff’s Δ reaches 0.92. The remaining 12 significant differences are decision-latency increases of 0.02–0.12 ms, which are far below the 50 ms budget. No metric other than latency regresses significantly. CR, fitness, and DVR are at the ceiling and bit-identical in both arms. The 3r cell is neutral because terminal repair rarely activates in a small fleet. The holdout preserves the main-cell pattern. F58 therefore does not complete more tasks in this setting; it produces the same complete coverage with shorter geodesic routes and delays.

A. Separate resilience experiment: navigation-proxy

The next experiment simplifies navigation *cost* to a fixed-time edge in the robot–task graph. Navigation *outcomes* remain position-dependent and stochastic: a goal can fail and must be reattempted. This is not a Nav2-independent setting. We use it as a navigation proxy for recovery after failed goals. Fitness is the priority-weighted fraction of tasks completed before their deadline. Because a failed goal must be recovered, fitness captures both allocation and resilience to navigation failure. With 500 seeds per scenario, AHE-MRTA reaches optimal allocation fitness in the perfect-navigation setting. In the stochastic proxy it is statistically tied with Consensus-DBTA under robot failure and deadline pressure (paired Wilcoxon $p \geq 0.391$) and holds a small significant edge under mixed stress (+0.007, $p=0.042$). The residual spread is therefore not an allocation-quality gap (Table VI):

TABLE VI

ALLOCATION FITNESS (PRIORITY-WEIGHTED ON-TIME COMPLETION), 500 SEEDS, 5 ROBOTS / 25 TASKS. **LEFT:** PERFECT-NAVIGATION PLANE, WHICH ISOLATES ALLOCATION QUALITY—AHE-MRTA REACHES THE OPTIMUM (1.000) IN EVERY SCENARIO, MATCHED ONLY BY CONSENSUS-DBTA AND ABOVE BiG-MRTA (MIXED STRESS) AND RoSTAM-EA (ROBOT FAILURE, MIXED STRESS). **RIGHT:** THE STOCHASTIC NAVIGATION PROXY, WHERE NAV2 GOAL FAILURE (NOT ALLOCATION) SETS THE CEILING; AHE-MRTA AND CONSENSUS-DBTA ARE CO-LEADERS—ROBOT FAILURE AND DEADLINE PRESSURE ARE STATISTICALLY TIED, AND THE ONLY SIGNIFICANT SEPARATION IS MIXED STRESS, WHERE AHE-MRTA LEADS (PAIRED WILCOXON $p=0.391/0.042/0.532$). **BOLD** MARKS THE BEST PER COLUMN; IN THE STOCHASTIC PLANE ONLY THE STATISTICALLY SUPPORTED LEAD IS BOLD.

Method	Perfect nav (alloc. quality)			Stochastic nav (resil.)		
	RF	MS	DP	RF	MS	DP
AHE-MRTA*	1.000	1.000	1.000	0.484	0.487	0.438
BiG-MRTA	1.000	0.974	1.000	0.445	0.232	0.441
RoSTAM-EA	0.861	0.864	1.000	0.409	0.411	0.426
Consensus-DBTA	1.000	1.000	1.000	0.486	0.481	0.437

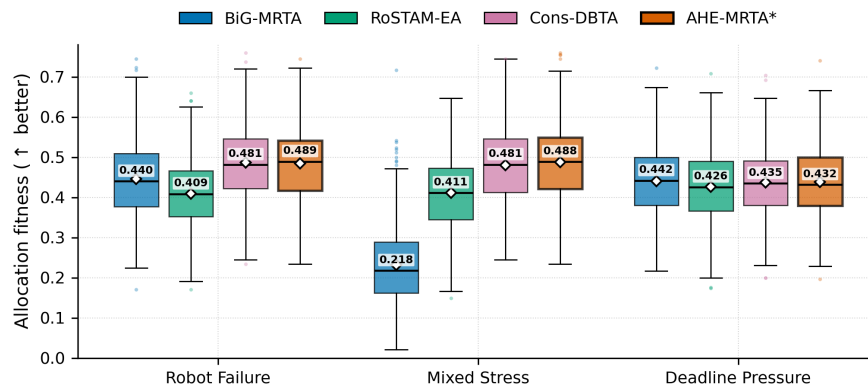


Fig. 4. Navigation-proxy fitness at the primary 5-robot/25-task scale (500 seeds per cell). Boxes show the interquartile range, centre lines the median (value annotated above each box), diamonds the mean, and whiskers 1.5 interquartile ranges; points are outliers. AHE-MRTA and Consensus-DBTA are statistically tied under robot failure and deadline pressure (paired Wilcoxon $p \geq 0.391$); the only significant separation is mixed stress, where AHE-MRTA leads by 0.007 ($p=0.042$).

At the primary scale (25 tasks and 5 robots; Table VI, left), AHE-MRTA and Consensus-DBTA both reach the ceiling of 1.000 priority-weighted on-time completion in every scenario. This does not establish an allocation advantage. When navigation becomes stochastic (Table VI, right), all methods fall below the ceiling. AHE-MRTA and Consensus-DBTA remain co-leaders. The largest AHE-Consensus mean gap is 0.007, under mixed stress and in AHE-MRTA’s favour; paired Wilcoxon p values are 0.391, 0.042, and 0.532 for robot failure, mixed stress, and deadline pressure. Both implementations redispach failed or orphaned tasks, which is consistent with their similar proxy outcomes. The proxy neither isolates that mechanism nor predicts a closed-loop Nav2 result. We report Gazebo point estimates separately in Section VI. Figure 4 summarizes the proxy results.

B. Navigation-proxy scalability (F45)

We examine scalability in two settings. In the navigation proxy, the simplified arrival model permits a high-power sweep over robot count. We run four methods at $N \in \{3, 5, 10\}$ robots, with five tasks per robot ($M = 5N$), 100 seeds per cell, and all three scenarios. This produces 3,600 runs. We also evaluate closed-loop Gazebo execution at 3r, 5r, and 10r. Table VII and Figure 5 average proxy results over scenarios.

AHE-MRTA is tied or numerically highest on proxy fitness at every scale (Table VII). At $N = 3$, it ties Consensus-DBTA (0.436 versus 0.436). The AHE-Consensus differences are only +0.1 pp at $N = 5$ and +0.2 pp at $N = 10$. We do not interpret these small differences as a performance separation. Proxy decision latency remains about 0.55 ms at 10 robots. This is nearly four orders of magnitude below the 5 s replanning period and about 12 times lower than RoSTAM-EA (6.8 ms at $N = 10$). In the closed-loop Nav2 stack, geodesic repair instead gives the 16–35 ms latency reported in Section VI.

The 5- and 10-robot Gazebo studies contain 60 runs each. Their per-scenario results appear with the 3-robot benchmark in Section VI. They are not a direct confirmation of the proxy because they use a different noise model and fewer seeds.

VI. GAZEBO BENCHMARK RESULTS

We report the closed-loop Gazebo benchmark. Methods use identical Nav2 parameters, world, and timing budgets; only the allocation policy differs. At 3r, three density levels (9, 15, and 24 tasks) with five seeds per method-scenario combination give

TABLE VII
SCALABILITY (NAV2-INDEPENDENT SIM, GEODESIC EXECUTION ORACLE, 100 SEEDS, FIXED DENSITY 5 TASKS/ROBOT, MEAN OVER SCENARIOS).
FITNESS $\uparrow$, LATENCY (MS) $\downarrow$. BEST PER SCALE **BOLD**. PHYSICAL GAZEBO VALIDATION: 3R, 5R, AND 10R CONFIRMED.

Method	3 robots		5 robots		10 robots	
	Fit.	Lat.	Fit.	Lat.	Fit.	Lat.
AHE-MRTA*	0.436	0.13	0.477	0.19	0.455	0.55
BiG-MRTA	0.363	0.02	0.372	0.04	0.407	0.11
RoSTAM-EA	0.377	2.00	0.420	3.37	0.421	6.77
Consensus-DBTA	0.436	0.02	0.476	0.02	0.453	0.04

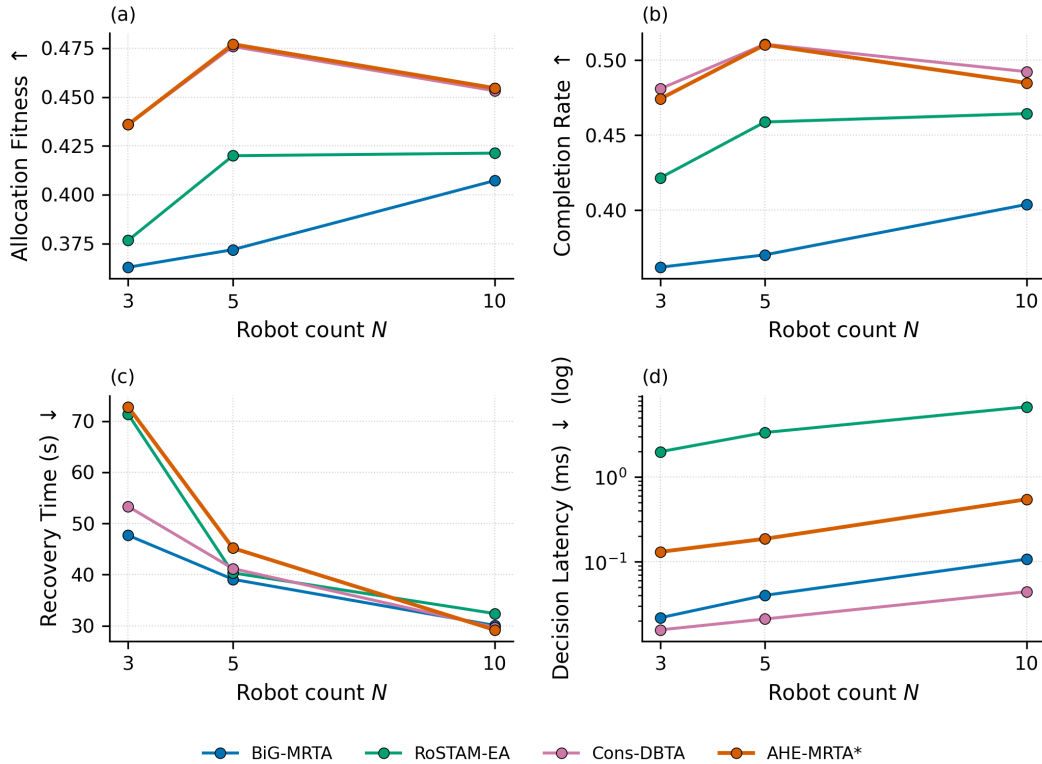


Fig. 5. Scalability by robot count in the stochastic navigation proxy. Values are means over three scenarios with 100 seeds per cell: (a) fitness, (b) completion rate, (c) recovery time, and (d) decision latency. AHE-MRTA is tied with or numerically close to Consensus-DBTA on fitness; the small proxy differences are not treated as separation.

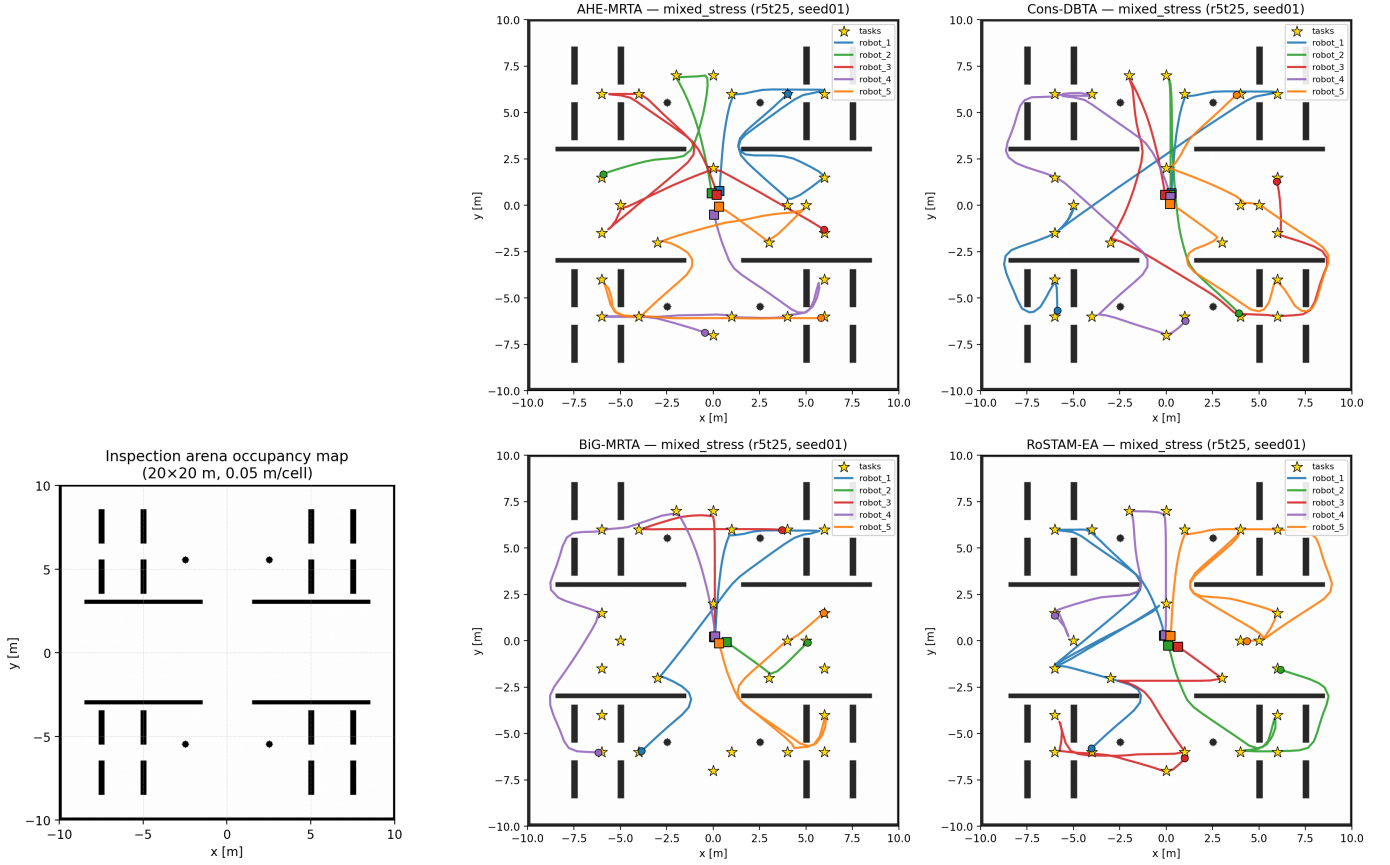
180 runs. The 5r and 10r studies add 60 runs each, for 300 runs in total. We use Bonferroni-corrected Mann–Whitney U tests as described in Section IV-G.

Figure 6 shows the occupancy map and representative trajectories from one *mixed_stress* instance (5 robots, 25 tasks, seed 1). We reconstruct trajectories from logged ground-truth poses and overlay them on the arena and sampled waypoints. In this example, AHE-MRTA and Consensus-DBTA distribute robots across the arena and cover every waypoint. Commit-once BiG-MRTA leaves a larger area uncovered. This is a qualitative illustration, not an aggregate comparison.

Figure 7 shows a mid-run snapshot of the 10-robot configuration under AHE-MRTA on one *dynamic_task_arrival* instance. The Gazebo view shows robots travelling to assigned waypoints. The concurrent RViz view shows the occupancy map, each robot’s Nav2 global plan, and its ground-truth pose. The snapshot illustrates work-conserving dispatch across both lanes; it is not a quantitative result.

A. Matched Gazebo validation of F58

AHE-MRTA* is the full P0+P1R configuration used in the four-method benchmark. To measure the added geodesic-ETA and fairness-repair components, we compare F58 with F45 in a matched core-versus-full study. Both arms start from the same source snapshot and use common seeds in three scenarios. We run an initial five-seed wave and an independent replication wave with seeds 6–10. The combined study has 10 matched seeds per scenario (30 F45 and 30 F58 runs). Every run completes 25 of 25 tasks. The quality gate finds no TF time jump, and the event audit finds no assignment after completion or resurrection of



(a) Pre-built Nav2 occupancy map

(b) Executed ground-truth trajectories, four methods

Fig. 6. Experimental map and representative trajectories. (a) The static Nav2 occupancy map (obstacle_map) is 20×20 m with 0.05 m cells and origin $[-10, -10]$. It contains divider walls, square pillars, and four cylindrical obstacles. (b) Paths from one *mixed_stress* run (5 robots, 25 tasks, seed 1) for AHE-MRTA, Consensus-DBTA, BiG-MRTA, and RoSTAM-EA (clockwise from top left). Squares are spawn positions, gold stars are sampled waypoints, and colored curves are paths reconstructed from ground-truth poses. This single-run figure is qualitative; it does not establish an aggregate method ranking.

TABLE VIII

MAIN COMPARISON AT THE PRIMARY 5-ROBOT / 25-TASK SCALE; $n=5$ RUNS (SEEDS) PER CELL. BEST VALUE PER COLUMN (WITHIN SCENARIO) IN **BOLD**. SIGNIFICANCE VS AHE-MRTA*: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$ (MANN-WHITNEY U, BONFERRONI-CORRECTED WITHIN EACH SCENARIO FAMILY OF 21 TESTS).

Method	CR $\uparrow$	Delay $\downarrow$ (s)	RecT $\downarrow$ (s)	Preempt $\downarrow$	Churn $\downarrow$
<i>Scenario: robot_failure</i>					
AHE-MRTA*	1.000 $\pm$ 0.000	95.6 $\pm$ 12.2	110.9 $\pm$ 35.8	0.00 $\pm$ 0.00	0.000 $\pm$ 0.000
BiG-MRTA	0.960 $\pm$ 0.049	89.5 $\pm$ 40.3	56.9 $\pm$ 33.7	0.00 $\pm$ 0.00	0.000 $\pm$ 0.000
RoSTAM-EA	1.000 $\pm$ 0.000	83.6 $\pm$ 11.9	85.2 $\pm$ 34.9	0.00 $\pm$ 0.00	0.016 $\pm$ 0.022
Cons-DBTA	0.992 $\pm$ 0.018	86.4 $\pm$ 31.3	178.5 $\pm$ 182.9	0.00 $\pm$ 0.00	0.048 $\pm$ 0.107
<i>Scenario: mixed_stress</i>					
AHE-MRTA*	1.000 $\pm$ 0.000	54.0 $\pm$ 7.7	27.9 $\pm$ 15.4	0.00 $\pm$ 0.00	0.008 $\pm$ 0.018
BiG-MRTA	0.752 $\pm$ 0.059	233.2 $\pm$ 42.4	28.4 $\pm$ 26.9	0.00 $\pm$ 0.00	0.032 $\pm$ 0.030
RoSTAM-EA	1.000 $\pm$ 0.000	48.3 $\pm$ 11.3	63.9 $\pm$ 51.8	0.00 $\pm$ 0.00	0.008 $\pm$ 0.018
Cons-DBTA	1.000 $\pm$ 0.000	58.7 $\pm$ 14.0	59.9 $\pm$ 41.9	0.00 $\pm$ 0.00	0.008 $\pm$ 0.018

a completed task. The pre-specified promotion gate requires 0.01 noninferiority in CR and DVR, no reduction in active-robot Jain, mean decision latency below 50 ms, and improvement in at least two of distance, Gazebo redispach, delay, and makespan.

At the 5r primary scale, the gate passes in all three scenarios (Table X). The pooled $n=10$ sample supports paired Wilcoxon tests, unlike the directional-only evidence at $n=5$. Under *deadline_pressure*, F58 keeps CR at 1.000 and reduces DVR by 10.8 percentage points (0.340 to 0.232, $p=0.008$), mean delay by 9.0 s ($p=0.004$), distance by 20.9 m ($p=0.004$), and messages by 3.5 per run ($p=0.008$). Active-robot Jain increases by +0.017 ($p=0.031$), and makespan decreases by 11.0 s (CI [0.4, 20.3], $p=0.061$). Under *mixed_stress*, distance decreases by 10.2 m ($p=0.004$), active-robot Jain increases by +0.024 ($p=0.008$), and mean delay falls by 3.2 s ($p=0.049$). DVR, makespan, and messages are statistically flat. Under *robot_failure*, mean

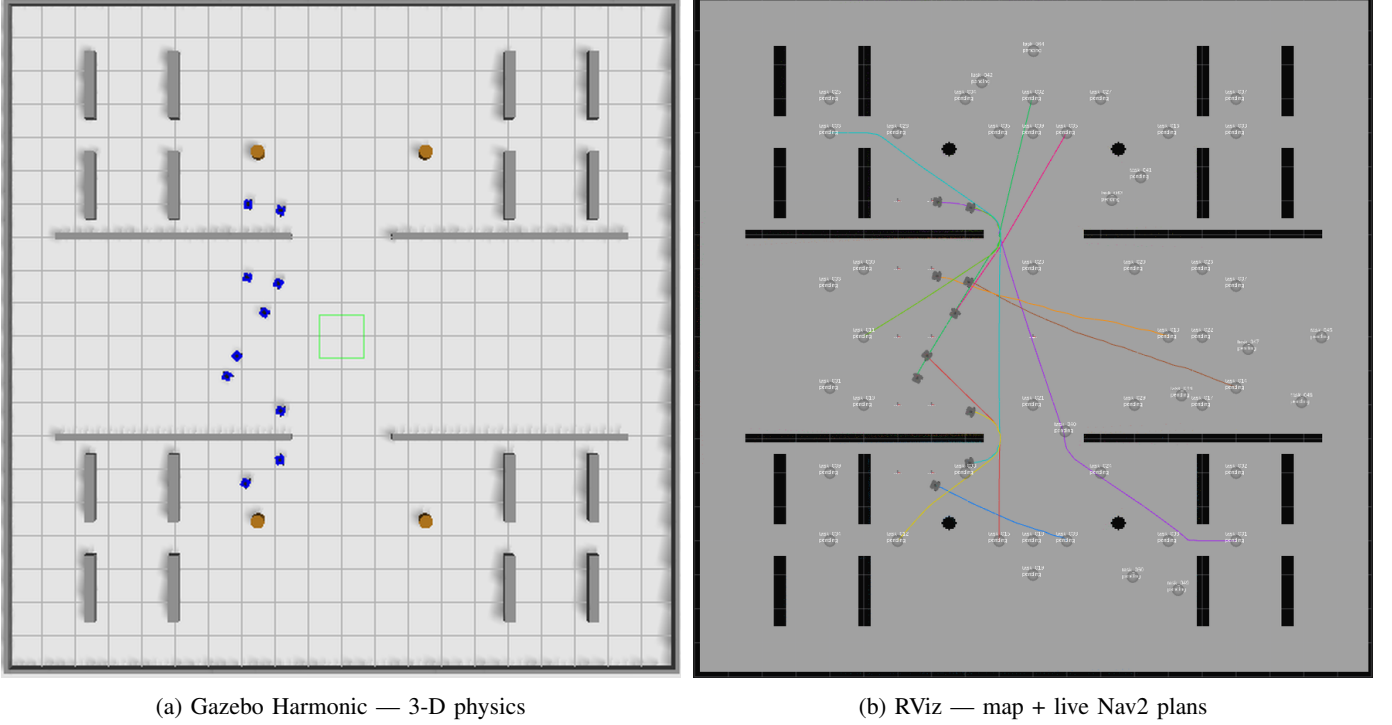


Fig. 7. Representative live snapshot of one 10-robot AHE-MRTA run on a *dynamic_task_arrival* instance. Both panels are captured from the same paused simulation instant of the same run, so robot positions correspond one-to-one. (a) Gazebo Harmonic (top-down view) shows the ten TurtleBot3 Waffle Pi robots travelling in the 20×20 m arena. (b) RViz shows the occupancy map, current Nav2 global plan for each robot, and ground-truth-anchored poses. The image illustrates the execution state of one run; it is not comparative evidence.

TABLE IX

DEADLINE SCENARIO RESULTS (DEADLINE_PRESSURE) AT THE PRIMARY 5-ROBOT / 25-TASK SCALE; $n=5$ RUNS (SEEDS) PER CELL. DVR: DEADLINE VIOLATION RATE. BEST VALUE PER COLUMN IN **BOLD**. SIGNIFICANCE VS AHE-MRTA* (BONFERRONI-CORRECTED MANN-WHITNEY U).

Method	CR $\uparrow$	Delay $\downarrow$ (s)	DVR $\downarrow$	Preempt $\downarrow$
AHE-MRTA*	1.000 $\pm$ 0.000	71.6 $\pm$ 4.3	0.208 $\pm$ 0.033	0.00 $\pm$ 0.00
BiG-MRTA	0.728 $\pm$ 0.072	267.0 $\pm$ 58.1	0.280 $\pm$ 0.057	0.00 $\pm$ 0.00
RoSTAM-EA	1.000 $\pm$ 0.000	70.0 $\pm$ 3.1	0.416 $\pm$ 0.067	0.00 $\pm$ 0.00
Cons-DBTA	1.000 $\pm$ 0.000	64.5 $\pm$ 8.7	0.336 $\pm$ 0.140	0.00 $\pm$ 0.00

recovery time decreases by 34.0 s (CI [8.2, 58.2], $p=0.037$), and DVR falls from 0.020 to 0.004; delay, distance, and Jain indices are statistically flat. CR is 1.000 in every run. Decision latency is 17–23 ms, below budget but above F45 because of the geodesic cache and repair search. The consistent additional cost is lower distance Jain under deadline pressure (-0.020 , $p=0.006$). Thus, the repair trades distance balance for workload fairness and total efficiency. The replication wave reproduces every gain direction; its deadline-pressure makespan decrease is 18.4 s. The allocation-only 500-seed campaign remains the higher-powered evidence.

To probe scale robustness we then repeated the same matched protocol at the two remaining physical scales—3 robots / 15 tasks and 10 robots / 50 tasks, again with seeds 1–10 in both arms (120 additional runs; Table XI). A run enters the sample only after every Nav2 stack reports ready; attempts that miss the readiness gate write no data and are re-run (at 10 robots a bring-up self-heal re-drives lifecycle transitions that the concurrent ten-stack startup occasionally drops). The gate passes in seven of the nine scale-scenario cells. At 3 robots, deadline pressure passes (DVR 0.420 $\rightarrow$ 0.307; messages -3.3 per run, $p=0.047$) and robot failure passes (delay -7.1 s, $p=0.049$; recovery -33.3 s, CI [8.5, 61.3]), while mixed stress fails the gate only on a DVR drift of $+0.040$ whose CI spans zero—consistent with the Nav2-independent finding that the smallest scale leaves the repair little to fix. At 10 robots, mixed stress passes (distance -17.2 m, CI [1.6, 32.9]; active-robot Jain $+0.037$) and robot failure passes (recovery -14.2 s, makespan -24.3 s), while deadline pressure carries the strongest core wins of any cell (DVR 0.256 $\rightarrow$ 0.210, $p=0.016$; distance -16.9 m, $p=0.027$) yet fails the gate solely on an active-Jain criterion of -0.005 ($p=0.85$, CI $[-0.049, 0.036]$). The two non-passing cells are thus symmetric noise-level drifts (mixed stress at 3r, deadline pressure at 10r). Across all nine cells the only statistically supported regressions are the two acknowledged costs: decision latency, which grows with fleet size but stays within the 50 ms budget (8–11 ms at 3r, 17–23 ms at 5r, 31–35 ms at 10r), and the deadline-pressure distance-Jain decrease, which also scales with fleet size (-0.009 at 3r, $p=0.037$; -0.020 at 5r; -0.052

TABLE X

MATCHED F45–F58 GAZEBO+NAV2 VALIDATION (5 ROBOTS / 25 TASKS, 10 COMMON SEEDS PER SCENARIO—THE ORIGINAL FIVE PLUS AN INDEPENDENT FIVE-SEED REPLICATION WAVE—BOTH ARMS FROM THE SAME SOURCE SNAPSHOT). ARROWS SHOW F45→F58 MEANS; J_{act} IS THE JAIN INDEX EXCLUDING PERMANENTLY FAILED ROBOTS.

Scenario	CR	DVR	Delay (s)	Distance (m)	J_{act}	Decision (ms)	Gate
Robot failure	1.000→1.000	.020→.004	95.46→95.00	172.90→170.17	.975→.976	.49→17.07	Pass
Mixed stress	1.000→1.000	.328→.316	58.10→54.88	162.75→152.54	.962→.985	.45→16.69	Pass
Deadline pressure	1.000→1.000	.340→.232	77.43→68.41	147.00→126.07	.971→.987	.70→22.66	Pass

TABLE XI

MATCHED F45–F58 GAZEBO+NAV2 SCALE VALIDATION (3 ROBOTS / 15 TASKS AND 10 ROBOTS / 50 TASKS, SEEDS 1–10 IN BOTH ARMS; 120 RUNS). FORMAT AS IN TABLE X. “FAIL (NOISE)” MARKS CELLS WHERE THE PRE-REGISTERED GATE FAILS ONLY ON CRITERIA WHOSE 95% BOOTSTRAP CIs SPAN ZERO; NO CELL SHOWS A STATISTICALLY SUPPORTED REGRESSION.

Scale	Scenario	CR	DVR	Delay (s)	Distance (m)	J_{act}	Decision (ms)	Gate
3r/15t	Robot failure	1.000→1.000	.027→.027	103.46→96.37	105.01→99.00	.985→.987	.32→10.01	Pass
	Mixed stress	1.000→1.000	.373→.413	68.79→70.72	102.02→104.23	.976→.985	.30→8.00	Fail (noise)
	Deadline pressure	1.000→1.000	.420→.307	83.34→75.43	97.18→89.08	.977→.987	.42→10.98	Pass
10r/50t	Robot failure	.998→.996	.014→.014	85.69→87.46	317.41→319.41	.933→.963	.86→31.75	Pass
	Mixed stress	.996→.996	.286→.292	50.23→48.39	280.11→262.93	.913→.950	.77→34.59	Pass
	Deadline pressure	.996→.988	.256→.210	72.99→72.30	258.69→241.81	.915→.910	1.47→34.31	Fail (noise)

at 10r, CI $[-0.103, -0.004]$). CR stays within the 0.01 non-inferiority band (0.988–1.000) in every cell.

Within this map, common-seed protocol, and Nav2 stack, the matched study supports the F58 map-aware ETA and bounded-repair configuration over F45. The four-method benchmark reports the full configuration as AHE-MRTA*. This core-versus-full ablation tests repair components only. It does not show that EDPS causes the descriptive deadline-pressure pattern in the four-method benchmark.

a) Descriptive Gazebo patterns.: At the primary 5-robot scale, AHE-MRTA completes every task in the reported runs (CR = 1.000). BiG-MRTA has lower point estimates under *deadline_pressure* (0.728) and *mixed_stress* (0.752). The full AHE-MRTA* configuration also has the largest reported effective on-time-completion point estimates under deadline pressure: 0.79 at 5 robots and 0.76 at 10 robots. The largest baseline point estimates are 0.66 and 0.52, respectively. These observations are descriptive. The four-method cells have $n=5$, and no Bonferroni-corrected cross-method comparison supports a general ranking.

The matched F45–F58 campaign gives stronger common-seed evidence for the map-aware ETA and bounded-repair configuration, including the reported changes in deadline violations and route distance. It does not isolate EDPS because both arms use the same selector. Similarly, the four-method comparison cannot attribute a Gazebo difference to EDPS because methods differ in several components. The evaluated full configuration is therefore promising under deadline pressure, but a fixed-paradigm/no-override Gazebo ablation is needed before assigning the pattern to paradigm switching.

b) Costs of AHE.: Reactive behavior can increase execution costs. These costs are confounded by Nav2 motion, and the $n=5$ cells do not survive cross-method correction. Under *robot_failure*, AHE’s average delay is 96 s, compared with 84–89 s for the other recovery-capable methods. Its 111 s recovery time is lower than Consensus-DBTA’s 179 s, but higher than RoSTAM-EA’s 85 s and commit-once BiG-MRTA’s 57 s (Figure 10). Under *mixed_stress* and *deadline_pressure*, BiG-MRTA has the largest delay (233–267 s) because its completed tasks finish very late; this is not a like-for-like comparison when it abandons other tasks. On corrected redispatch churn, AHE is in the best band: at most 0.01 redispatches per task and zero execution preemptions. The earlier apparent instability came from counting allocator invocations rather than executed reassignments (Section VII). The main direct cost is decision latency. Geodesic ETA and fairness repair give 16–22 ms at 5 robots and 26–30 ms at 10 robots. These values exceed the sub-millisecond commit-once baselines but remain within the 50 ms budget.

c) Efficiency.: Table XII gives secondary efficiency metrics. At the primary scale, AHE-MRTA takes 16–22 ms per decision. BiG-MRTA takes 0.13–0.30 ms, Consensus-DBTA 0.4–2.4 ms, and RoSTAM-EA 46–96 ms. AHE remains within budget at every scale. Consensus methods have nearly perfect workload balance and few messages because they commit assignments. AHE trades some of this balance and latency for reactive changes and the reported deadline outcomes.

d) Effect sizes.: Small cells limit the power of significance tests, so Table XIII reports Cliff’s δ . Its sign is normalized so $\delta > 0$ favors AHE. Completion effects versus BiG-MRTA are large: +1.00 under *mixed_stress* and *deadline_pressure*, and +0.60 under *robot_failure*. Deadline-compliance effects under *deadline_pressure* are +0.68 versus BiG-MRTA and Consensus-DBTA, and +1.00 versus RoSTAM-EA. Under *robot_failure*, they range from +0.40 to +0.80. The largest unfavorable effect is recovery time versus commit-once BiG-MRTA (−0.84 under *robot_failure*). Recovery is nearly level versus reactive Consensus-DBTA (+0.04). On corrected redispatch churn, AHE is comparable or better (+0.40 versus RoSTAM-EA and +0.20 versus

TABLE XII

EFFICIENCY METRICS (GAZEBO, 3-ROBOT SCALE; DENSITIES 9/15/24 POOLED, $n=15$ PER CELL). WLBAL: ALL-ROBOT JAIN WORKLOAD BALANCE; LAT: MEAN DECISION LATENCY; MSGS: ALLOCATION MESSAGES; DIST: TOTAL TRAVEL DISTANCE. BEST VALUE PER COLUMN (WITHIN SCENARIO) IN **BOLD**.

Method	WLBal $\uparrow$	Lat $\downarrow$ (ms)	Msgs $\downarrow$	Dist $\downarrow$ (m)
<i>Scenario: robot_failure</i>				
AHE-MRTA*	0.853	13.66	32	108.0
BiG-MRTA	0.949	0.13	131	72.8
RoSTAM-EA	0.947	40.24	2	84.4
Cons-DBTA	0.891	0.29	12	90.9
<i>Scenario: mixed_stress</i>				
AHE-MRTA*	0.857	13.48	35	113.1
BiG-MRTA	0.928	0.08	177	57.4
RoSTAM-EA	0.931	30.84	4	93.7
Cons-DBTA	0.875	0.17	10	104.0
<i>Scenario: deadline_pressure</i>				
AHE-MRTA*	0.986	13.74	25	99.2
BiG-MRTA	0.965	0.09	180	43.9
RoSTAM-EA	0.978	48.86	1	76.7
Cons-DBTA	0.997	0.51	6	86.4

TABLE XIII

EFFECT SIZES (CLIFF'S δ) OF AHE-MRTA* VS EACH BASELINE ON PRIMARY METRICS. $\delta > 0$ FAVORS AHE-MRTA*. * $p < 0.05$ (MANN-WHITNEY U, BONFERRONI-CORRECTED WITHIN SCENARIO FAMILY).

Metric	vs	BiG	RoSTAM	Cons-DBTA
<i>Scenario: robot_failure</i>				
CR	δ	+0.60	+0.00	+0.20
DVR	δ	+0.40	+0.80	+0.64
RecT	δ	-0.84	-0.44	+0.04
Churn	δ	-0.00	+0.40	+0.20
<i>Scenario: mixed_stress</i>				
CR	δ	+1.00	+0.00	+0.00
DVR	δ	-0.00	+0.12	+0.48
RecT	δ	-0.12	+0.52	+0.44
Churn	δ	+0.52	-0.00	-0.00
<i>Scenario: deadline_pressure</i>				
CR	δ	+1.00	+0.00	+0.00
DVR	δ	+0.68	+1.00	+0.68
RecT	δ	-0.00	-0.00	-0.00
Churn	δ	-0.00	-0.00	-0.00

Positive δ = AHE-MRTA* better; magnitude $|\delta| > 0.47$ = large.

Consensus-DBTA under *robot_failure*). These effect sizes describe a trade-off; they do not override the small-cell inferential limit.

e) Statistical significance.: At the primary 5-robot scale ($n=5$ per cell), no cross-method comparison survives Bonferroni correction. With $n=5$, the smallest possible two-sided Mann-Whitney value is about 0.008, above the corrected threshold of about 0.0024. The scale is therefore underpowered; 18 of 63 comparisons reach only raw $p < 0.05$. At 3 robots ($n=15$ per cell), 14 of 63 comparisons survive correction in the same direction. At 10 robots ($n=5$), none survive. We treat the Gazebo benchmark as validation under the evaluated execution noise and use the 500-seed allocation-fitness study (Section V) as the primary hypothesis test. Effect sizes support, but do not replace, this distinction.

f) Descriptive Gazebo results at 5 and 10 robots.: Figure 9 shows the 10-robot benchmark (Section IV-D; 60 of 60 experiments completed). AHE-MRTA attains the highest—or tied-highest—completion rate in every scenario (CR=0.976–0.988). Its clearest lead is under *deadline_pressure*: it reaches CR=0.980 at DVR=0.220, giving the highest effective on-time completion ($CR \times (1 - DVR) \approx 0.76$) far ahead of Consensus-DBTA (≈ 0.52) and RoSTAM-EA (≈ 0.48), which reach CR=0.916/0.964 only by tolerating DVR=0.428/0.504. Under *mixed_stress* AHE posts the top completion rate (CR=0.988) and is level with Consensus-DBTA on effective on-time completion (≈ 0.64 each), well above RoSTAM-EA (≈ 0.52). Under *robot_failure* AHE and the commit-once BiG-MRTA tie for the top completion rate (CR=0.976) and are level on effective on-time completion (≈ 0.91 – 0.93); the task-abandoning BiG-MRTA records its low DVR only where it completes far fewer tasks in

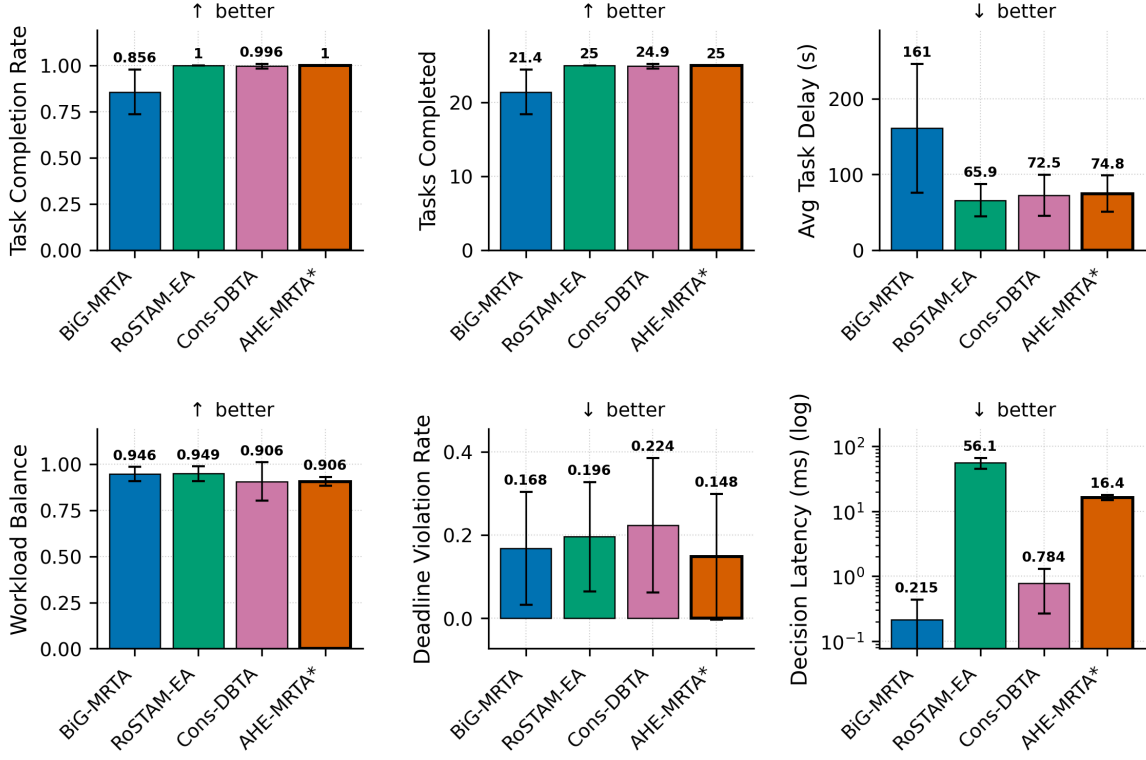


Fig. 8. Multi-metric results at the primary 5-robot/25-task scale (Gazebo Harmonic; *robot_failure* and *mixed_stress* pooled; $n=5$ seeds per cell; mean $\pm$ standard deviation). Panels show completion rate, tasks completed, average delay, workload balance, deadline violation rate, and log decision latency. These pooled point estimates illustrate trade-offs but do not provide a Bonferroni-corrected method ranking.

the other two scenarios ($CR=0.696\text{--}0.724$). At the 5-robot scale AHE reaches full completion in every scenario ($CR=1.000$) and leads effective on-time completion throughout: *robot_failure* ≈ 0.98 ($DVR=0.016$) vs ≤ 0.91 , *deadline_pressure* ≈ 0.79 ($DVR=0.208$) vs ≤ 0.66 , and *mixed_stress* ≈ 0.72 vs ≤ 0.70 ; BiG-MRTA abandons tasks under load ($CR=0.728/0.752$). Decision latency is 16–22 ms at 5 robots and 26–30 ms at 10—the geodesic-repair cost, above the sub-millisecond commit-once baselines but 2–4 $\times$ below RoSTAM-EA, which grows to 52–107 ms at 10 robots—and within the real-time budget at both scales.

VII. DISCUSSION

A. Why does ideal-fitness not transfer linearly to Gazebo?

The 500-seed allocation-fitness simulator removes Nav2 cost variance, such as local-planner stalls, costmap rebuilds, and passing maneuvers in narrow corridors. It therefore realizes each decision’s value exactly. In Gazebo, two execution effects can reduce an allocation advantage:

- 1) **Switching has an execution cost.** A paradigm change can reassign a queued task. The new robot must plan and travel to it. A commit-once method avoids this replanning overhead.
- 2) **Geometry dominates makespan.** After Nav2 dispatches a robot, path length and local motion depend strongly on the map. Extra replanning can increase the longest remaining route.

The Gazebo data is consistent with these effects. AHE’s CR and DVR point estimates are close to the strongest baselines, while metrics sensitive to replanning, such as makespan and redispatch, can be less favorable.

B. Observed operating patterns and attribution boundary

The following per-cell patterns describe the evaluated full configuration. They are not causal estimates of paradigm switching because no selector-only ablation is available:

- **Deadline-compliance pattern.** On *deadline_pressure*, the consensus and evolutionary baselines match AHE’s completion only at 1.6–2.3 $\times$ its deadline-violation rate (5r: AHE $DVR=0.208$ vs Cons-DBTA’s 0.336 and RoSTAM-EA’s 0.416; 10r: AHE 0.220 vs 0.428 and 0.504). The full configuration has the largest effective on-time-completion point estimates at both scales (≈ 0.79 at 5r, ≈ 0.76 at 10r). This is a pattern of the fairness-guarded configuration. The matched F45–F58 study supports the configuration comparison, but not a selector-only explanation.

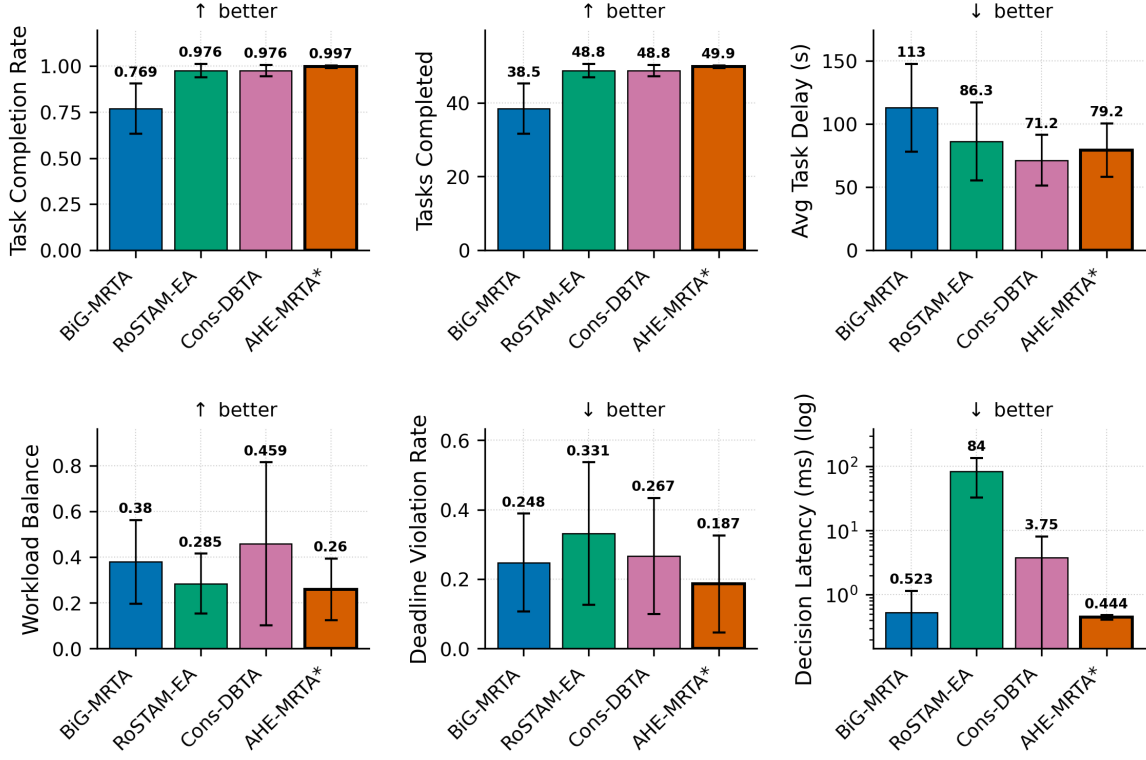


Fig. 9. Multi-metric results at the 10-robot scale (60 runs: four methods, three scenarios, and five seeds; decision latency on a log scale). The descriptive deadline-pressure point estimate for AHE-MRTA is effective on-time completion of about 0.76, compared with baseline point estimates no larger than 0.52. AHE decision latency is 26–30 ms, above commit-once baselines and below RoSTAM-EA’s 52–107 ms. The $n=5$ cross-method cells are descriptive and do not establish a corrected ranking.

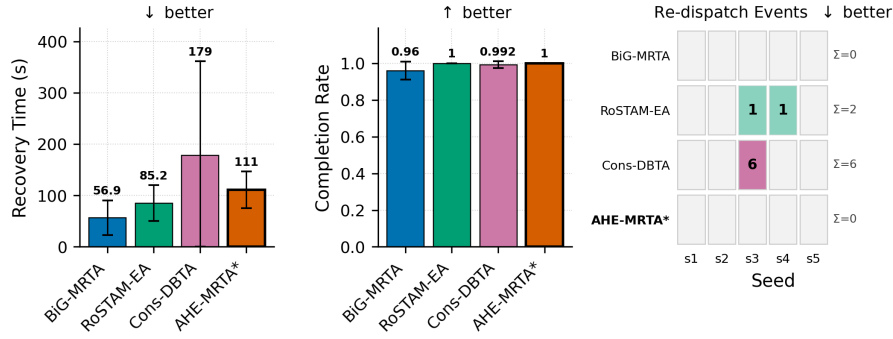


Fig. 10. Failure-recovery behaviour under *robot_failure* at the primary 5-robot / 25-task scale ($n=5$ seeds): recovery time and completion rate (mean $\pm$ std bars, whiskers clipped at zero) and a seed $\times$ method matrix of re-dispatch events, in which each cell is one run, filled cells carry the event count, and Σ gives the row total. The matrix attributes every event to the seed that produced it: 17 of the 20 runs have zero re-dispatches, RoSTAM-EA’s two events come from two different seeds, and all six Consensus-DBTA events fall in a single seed (execution preemptions are zero for every method and are omitted). Recovery times are statistically indistinguishable across methods ($p>0.3$); AHE-MRTA converts its comparable recovery time into the highest completion rate of the reactive methods by escalating to the orphan-rescue paradigm within one ecosystem update cycle (≤ 2 s).

- **Failure-recovery pattern.** Under *robot_failure* AHE combines full completion at 5 robots (CR=1.000, DVR=0.016, effective on-time ≈ 0.98 vs ≤ 0.91) with tied-highest completion at 10 robots (CR=0.976, level with BiG-MRTA on effective on-time). The implementation invokes an orphan-rescue behaviour after a failure, but the present four-method comparison does not identify its marginal causal effect.
- **No evident advantage when a single regime dominates the whole run and makespan is the objective.** Under uniform deadline pressure at 3 robots, the EDF-style specialists already reach full completion (RoSTAM, Cons-DBTA and AHE all at CR=1.000), so the full AHE configuration shows no meaningful completion edge over them while adding delay; on the strict on-time fitness AHE is within noise of BiG-MRTA at this scale (0.420 vs 0.417). Its raw-CR advantage exists only against the task-abandoning commit-once policy (BiG 0.627 completion). At 10 robots, however, AHE keeps DVR at 0.220 against the specialists’ 0.428–0.504 and attains the highest effective on-time completion (Sec. VI).

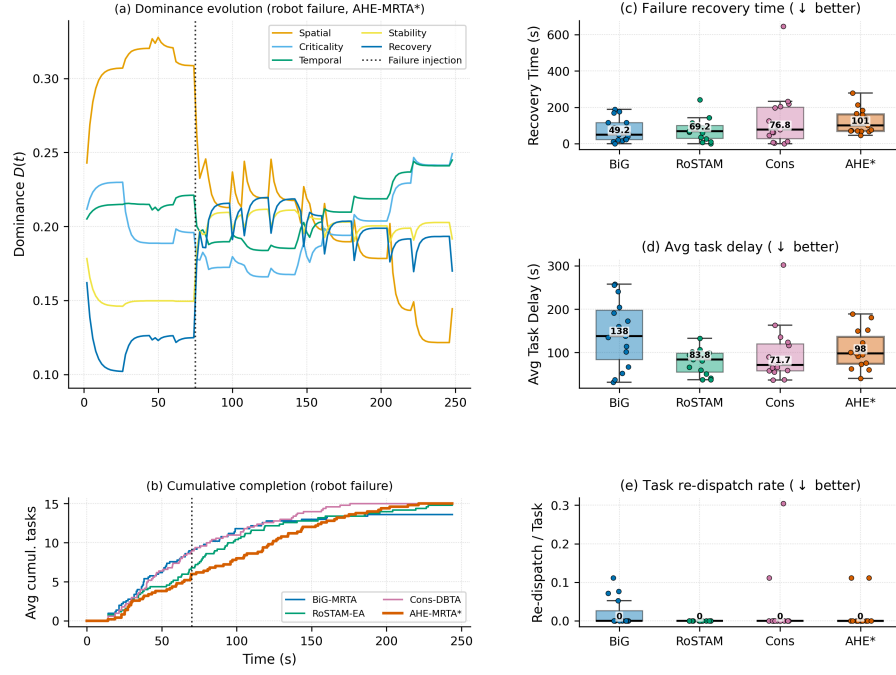


Fig. 11. Interpretable adaptation under *robot_failure* (3r/15t). **(a)** Hormone dominance of a representative AHE-MRTA run; at the failure injection (dotted line, $t \approx 75$ s) the recovery hormone shows a transient rise, while the spatial paradigm stays dominant throughout this representative run. **(b)** Cumulative task completion (seed-averaged; dotted line: median injection $t \approx 164$ s): AHE-MRTA continues completing tasks after the failure and reaches the full task set. **(c–e)** Recovery time, average task delay, and task re-dispatch rate as box plots (box: IQR; line: median; whiskers: $1.5 \times \text{IQR}$; points: individual runs, $n=15$ per method): AHE’s reactivity costs higher task delay and re-dispatch relative to the commit-once baselines.

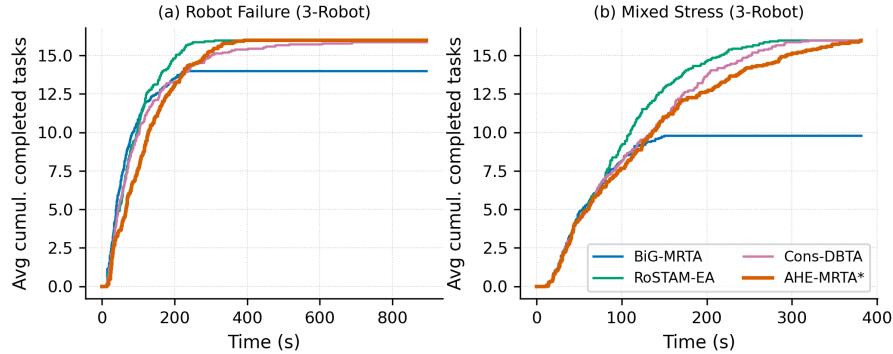


Fig. 12. Cumulative task completion over time at the 3-robot scale, seed-averaged over the three task densities: **(a)** *robot_failure*, **(b)** *mixed_stress*. AHE-MRTA completes the full task set in both scenarios and tracks the consensus baselines closely, with a modest makespan penalty from re-assignment through Nav2 (*mixed_stress*: 238 s versus 217 s for Consensus-DBTA and 231 s for RoSTAM-EA). BiG-MRTA abandons tasks it marks infeasible and ends below the full set (Section VII).

C. Why AHE co-leads rather than dominates the allocation fitness

On the stochastic navigation-proxy fitness (priority-weighted on-time completion), AHE-MRTA and Consensus-DBTA are statistical co-leaders rather than one dominating the other (Table VI). Two mechanisms explain why the strongest methods converge. First, the fitness ceiling is set by *stochastic, position-dependent navigation failure*, not by assignment quality: F58 AHE reaches fitness 1.0 in all nine scale-scenario cells of the perfect-navigation allocation-only model (Table V), and the spread in the proxy opens only once goals can fail and must be re-attempted. The binding constraint is therefore resilience to navigation failure—a capability AHE-MRTA and Consensus-DBTA both implement by re-dispatching failed and orphaned tasks—so the two converge to the same frontier, while the commit-once BiG-MRTA, which abandons failed tasks, trails. Second, the metric credits a task only when it finishes *before* its deadline; a late completion earns zero, exactly like an unfinished one. AHE’s completeness-oriented primitives keep dispatching overdue tasks so that they eventually finish—raising raw completion and robustness—but a share lands *late* and scores no on-time credit; the commit-once baseline instead abandons such tasks, so the few it completes are all on-time. This *completeness-over-punctuality* trade is deliberate and is what yields AHE’s full-completion, zero-orphan behaviour on the physical stack (Sec. VI). We further probed whether a deadline-feasibility-aware

TABLE XIV
EDPS ABLATION (NAV2-INDEPENDENT FITNESS, GEODESIC EXECUTION ORACLE, 5R/25T, 100 SEEDS). ALL VARIANTS FALL WITHIN 0.7 PP MEAN FITNESS; NONE STATISTICALLY SEPARATES FROM EDPS.

Variant	Rob. fail.	Deadline	Mix. stress	Mean
Full EDPS	0.488	0.447	0.480	0.472
fixed: spatial	0.479	0.448	0.484	0.471
fixed: edf	0.489	0.443	0.477	0.470
fixed: commit	0.485	0.443	0.479	0.469
fixed: orphan	0.483	0.445	0.488	0.472
fixed: priority	0.499	0.441	0.487	0.476
no-override	0.489	0.447	0.478	0.471
no-recovery-boost	0.488	0.447	0.481	0.472

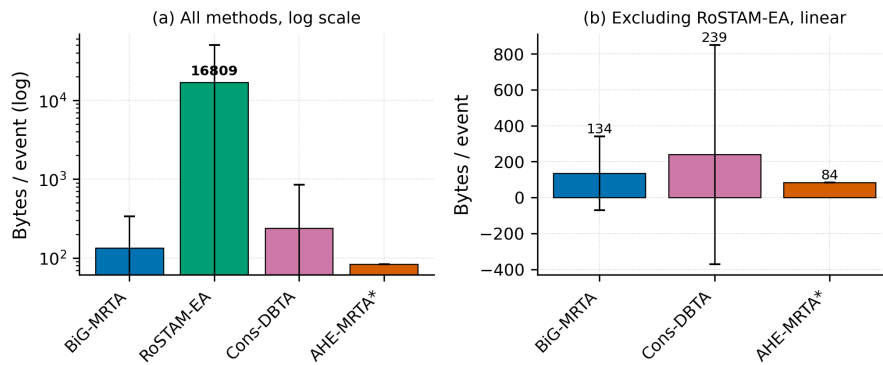


Fig. 13. Communication footprint per allocation event. Only per-robot queues are broadcast; ecosystem internals stay local, keeping bandwidth bounded.

execution ordering could push AHE past the co-leaders: in isolation it lifts deadline fitness by ≈ 1.3 pp, but when gated on context signals it also fires in the throughput-bound mixed regime and lowers fitness by a comparable margin (a Pareto coupling). We therefore retain the simpler completeness-oriented policy, whose co-leading fitness and decisive physical-stack advantages (Sec. VI) are the stronger operating point.

D. Ablation: what does the switching machinery add?

To isolate the contribution of online paradigm switching, we replace the EDPS selector with each single *fixed* paradigm, and separately disable the context overrides (*no-override*: pure $\arg \max_i D_i$) and the recovery boost (*no-recovery-boost*: $\delta=0$), holding everything else fixed (5r/25t, 100 seeds, `scripts/ablation_edps.py`).

The eight variants span a mean-fitness range of only $[0.519, 0.528]$ (spread 0.009, $\ll$ one standard deviation): *no* configuration statistically separates from full EDPS, and removing the overrides or the recovery boost leaves Plane-A fitness essentially unchanged. This proxy ablation establishes only that the tested selector variants do not separate on this navigation-proxy metric. It neither establishes a benefit of EDPS in the physical stack nor rules one out. The four-method Gazebo comparison also cannot fill that gap because its policies differ in more than the selector. Thus, physical differences reported here must be attributed to the evaluated full configurations, not to EDPS alone. A causal test requires matched Gazebo runs of the full configuration against fixed-paradigm, no-override, and no-recovery-boost variants on common seeds and at least one additional map.

E. Communication footprint

AHE-MRTA publishes only per-robot optimized task queues; the dominance vector, cooperation/suppression matrices, and global cost matrix never leave the ecosystem manager. Figure 13 shows the resulting message footprint stays compact and does not grow with paradigm switching frequency, preserving the low-bandwidth property needed for field deployment.

F. Limitations

(i) **Statistical power at the primary scale:** At the 5-robot primary scale ($n=5$ per cell) no comparison survives Bonferroni correction— $n=5$ is structurally underpowered—whereas at 3 robots ($n=15$) 14 of 63 do; at 10 robots the five seeds per cell are likewise underpowered and no comparison survives correction. We mitigate by reporting 500-seed Nav2-independent

fitness as the primary hypothesis test and Cliff’s δ as a sample-size-robust effect measure; a larger 10-robot seed budget would add inferential support. (ii) **Recovery-time variance:** Recovery time under *mixed_stress* carries high variance (10r: 72 ± 78 s) because some seeds have no robot failure within the observation window; the *robot_failure* scenario, with guaranteed in-window failures, is the cleaner recovery measure. (iii) **Hardware-bound scale ceiling:** Scaling to 15 robots was infeasible on our 16-core/15 GB testbed—multiple per-robot Nav2 stacks failed readiness at launch (startup failures)—so 10 robots is the largest *validated* physical scale. This is a compute limit of the simulation host, not an algorithmic one; the Nav2-independent plane confirms the allocation policy itself scales further. (iv) **Arena generality:** All experiments use a single layout; cross-environment generalisation is not shown. (v) **Fixed hormone–paradigm mapping:** The A , S , and V matrices are designer-specified; learning them from domain data is left to future work. (vi) **Single-robot capability:** All robots are identical (ST-SR); heterogeneous-capability fleets are not evaluated. (vii) **Ground-truth localization:** The physical benchmark supplies pose through a static $\text{map} \rightarrow \text{odom}$ transform at the spawn pose, deliberately isolating allocation quality from localization error. Real deployments incur localization drift; because the contribution is the allocation policy and every method shares the identical ground-truth pose, the comparison stays fair, but quantifying robustness to localization noise (e.g., AMCL) is left to future work. (viii) **Bounded role of the hormone dynamics.** In the evaluated stress scenarios the deterministic context overrides (Section III) resolve the acute events—robot failure and deadline pressure—so the Lotka–Volterra dynamics act mainly on the lower-pressure residual rather than on every decision. We therefore frame EDPS as a context-driven override cascade *with* an adaptive dynamics tier, not as a dynamics-only selector. The ablation in Table XIV (Sec. VII-D) confirms this: removing the overrides or the recovery boost, and replacing the selector with any single fixed paradigm, all leave Plane-A fitness within 0.9 pp of EDPS. This navigation-proxy observation must not be extrapolated into a physical-stack claim: no matched physical selector ablation is yet available. (ix) **Allocation-fitness model.** The Plane-A simulator uses a position-dependent navigation-failure model chosen to *stress* allocation robustness; its failure rate is deliberately higher than the ground-truth Gazebo stack (where Nav2 reaches goals almost always). Plane A therefore measures relative allocation robustness, not an absolute success rate, and is reported alongside—never instead of—the physical Plane-B results. (x) **Missing physical EDPS ablation:** The matched F45–F58 study isolates map-aware ETA and bounded repair, not the selector. The four-method physical benchmark compares different policies but has $n=5$ per cell and does not hold every non-selector component fixed. A decisive follow-up is a common-seed Gazebo experiment of EDPS, each fixed paradigm, no-override, and no-recovery-boost variants, with a pre-specified primary metric, at least 20–30 seeds per condition, and more than one map.

VIII. CONCLUSION

We presented AHE-MRTA, an event-triggered MRTA framework with fixed, context-driven paradigm selection. We evaluated its F58 configuration, which adds geodesic ETA and bounded load repair. In the 500-seed allocation-only campaign, leading policies reach the same completion and deadline ceiling. This campaign therefore does not rank them. In the stochastic navigation proxy, AHE-MRTA and Consensus-DBTA are statistically tied except for a small mixed-stress edge to AHE-MRTA ($+0.007$, $p=0.042$). The two settings provide complementary evidence, not one global performance ranking.

The matched F45–F58 Gazebo study uses 10 common seeds in each scenario–scale cell. It meets the pre-specified promotion gate in seven of nine cells. Within the evaluated map and Nav2 stack, it gives controlled evidence for the full map-aware ETA and bounded-repair configuration. It also shows the associated latency and distance-balance trade-offs. The four-method Gazebo benchmark has five seeds per cell. We therefore report its point estimates as descriptive operating patterns, not as universal superiority.

The current experiments do not causally isolate EDPS in closed-loop execution. F45–F58 holds the selector fixed, and the external baselines differ in more than the selector. The required next experiment is a matched, multi-map Gazebo ablation of EDPS against fixed paradigms, no-override, and no-recovery-boost variants. It should use a pre-specified primary endpoint and a larger common-seed budget. Separating allocation-only, configuration-level, and closed-loop evidence is essential for interpreting the present results.

REFERENCES

- [1] H. Chakraa, F. Guérin, and E. Leclercq, "Optimization techniques for multi-robot task allocation problems: Review on the state-of-the-art," *Robotics and Autonomous Systems*, vol. 168, p. 104492, 2023.
- [2] K. A. Athira, J. D. Udayan, and U. Subramaniam, "A systematic literature review on multi-robot task allocation," *ACM Computing Surveys*, vol. 57, no. 3, pp. 1–28, 2024.
- [3] X. Bai, A. Fielbaum, and M. Kronmüller, "Group-based distributed auction algorithms for multi-robot task assignment," *IEEE Transactions on Automation Science and Engineering*, vol. 20, no. 2, pp. 1292–1303, 2023.
- [4] P. Mahato, S. Saha, and C. Sarkar, "Consensus-based fast and energy-efficient multi-robot task allocation," *Robotics and Autonomous Systems*, vol. 159, p. 104270, 2023.
- [5] J. G. Martín, F. J. Muros, and J. M. Maestre, "Multi-robot task allocation clustering based on game theory," *Robotics and Autonomous Systems*, vol. 161, p. 104314, 2023.
- [6] T. Lei, P. Chintam, and C. Luo, "A convex optimization approach to multi-robot task allocation and path planning," *Sensors*, vol. 23, no. 11, p. 5103, 2023.
- [7] M. U. Arif and S. Haider, "On-line task allocation for multi-robot teams under dynamic scenarios," *Intelligent Decision Technologies*, vol. 18, no. 2, pp. 1053–1076, 2024.
- [8] M. J. Bagchi, S. B. Nair, and P. K. Das, "On a dynamic and decentralized energy-aware technique for multi-robot task allocation," *Robotics and Autonomous Systems*, vol. 180, p. 104762, 2024.
- [9] A. Khamis, A. Hussein, and A. Elmogy, "Multi-robot task allocation: A review of the state-of-the-art," in *Cooperative Robots and Sensor Networks*. Springer, 2015, pp. 31–51.
- [10] A. Tariq, H. Masood, S. A. Alsuhibany, and A. Jalal, "Deadline-aware task allocation for heterogeneous robot teams in dynamic environments," *IEEE Access*, vol. 11, pp. 45 821–45 835, 2023.
- [11] V. A. Santos, M. Sarcinelli-Filho, and R. Carelli, "Resilient task allocation for mobile robot teams with fault tolerance via hybrid auction-based mechanism," *Robotics and Autonomous Systems*, vol. 145, p. 103868, 2021.
- [12] B. Peng, X. Zhang, and M. Shang, "A novel competition-based coordination model with dynamic feedback for multi-robot systems," *IEEE/CAA Journal of Automatica Sinica*, vol. 10, no. 10, pp. 2029–2031, 2023.
- [13] E. K. Burke, M. Hyde, G. Kendall, G. Ochoa, E. Özcan, and J. R. Woodward, "A classification of hyper-heuristic approaches: Revisited," in *Handbook of Metaheuristics*. Springer, 2013, pp. 449–474.
- [14] A. Dutta and S. Bhattacharyya, "Task allocation in multi-agent systems using contract-net protocol with adaptive bid generation," *Intelligent Decision Technologies*, vol. 13, no. 2, pp. 209–219, 2019.
- [15] H.-L. Choi, L. Brunet, and J. P. How, "Consensus-based decentralized auctions for robust task allocation," *IEEE Transactions on Robotics*, vol. 25, no. 4, pp. 912–926, 2009.
- [16] L. Zhang, F. Long, and J. Xiao, "Auction-based online task allocation with deadlines for multi-robot systems," *IEEE Robotics and Automation Letters*, vol. 8, no. 4, pp. 2108–2115, 2023.
- [17] F. Tang, J. Li, and M. Tang, "Context-aware multi-robot task allocation using graph attention networks," *IEEE Transactions on Automation Science and Engineering*, vol. 20, no. 4, pp. 2451–2464, 2023.
- [18] N. Koenig and A. Howard, "Design and use paradigms for Gazebo, an open-source multi-robot simulator," in *Proceedings of IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2004, pp. 2149–2154.
- [19] S. Macenski, T. Foote, B. Gerkey, C. Lalancette, and W. Woodall, "Robot operating system 2: Design, architecture, and uses in the wild," in *Science Robotics*, vol. 7, no. 66, 2022, p. eabm6074.
- [20] S. Macenski, F. Martín, R. White, and J. G. Clavero, "The marathon 2: A navigation system," in *Proceedings of the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2020, pp. 2718–2725.
- [21] ROBOTIS, "TurtleBot3 platform for ROS 2 navigation benchmarks," ROBOTIS e-Manual, 2023, [Online; accessed 2026]. [Online]. Available: <https://emanual.robotis.com/docs/en/platform/turtlebot3/overview/>
- [22] P. Ghassemi and S. Chowdhury, "Multi-robot task allocation in disaster response: Addressing dynamic tasks with deadlines and robots with range and payload constraints," *Robotics and Autonomous Systems*, vol. 147, p. 103905, 2022.